K-FIELD DUAL-AXIS KINETIC MASS CMB TRAVERSAL LATERAL-FORCE COMPUTATION

Canonical formula kernel, exact 2:1 compound-rotation geometry, anisotropic traversal-cost force, lateral projection, cycle impulse, validation gates, and code-stack foundation

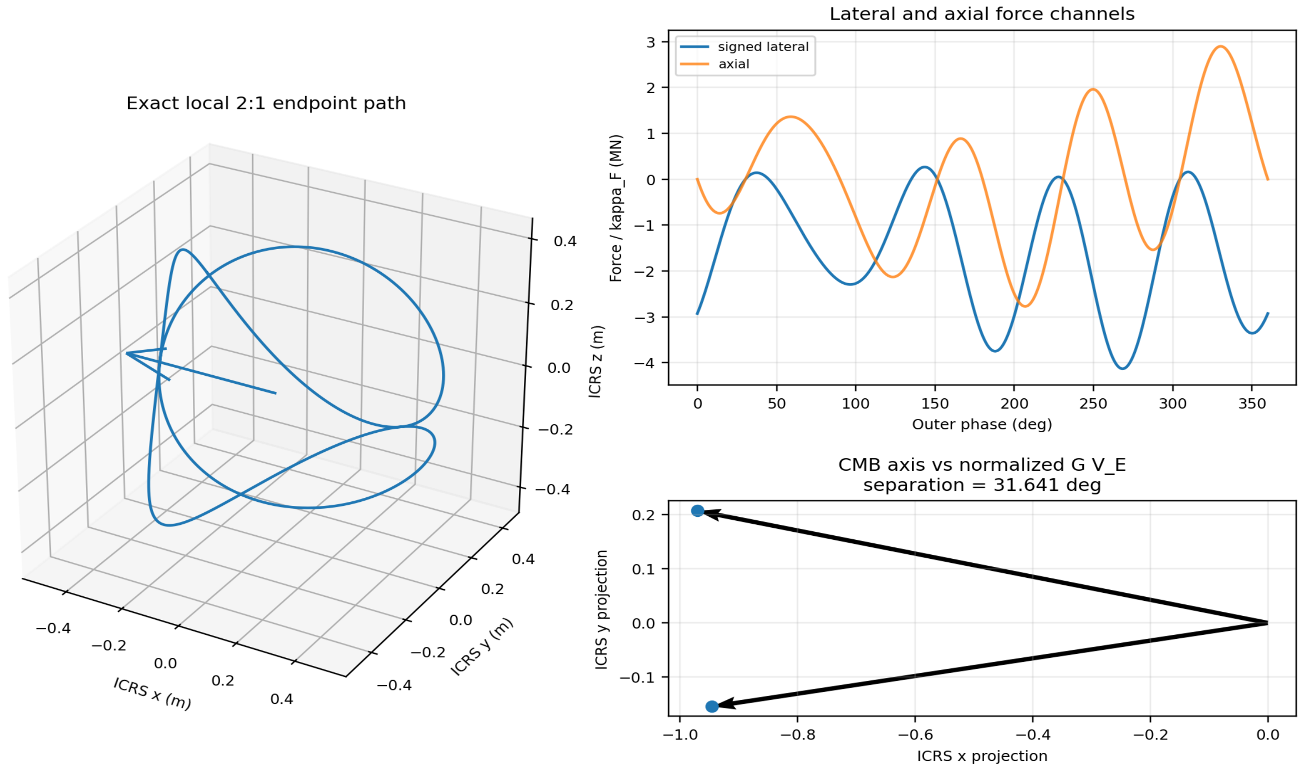

Figure 1. Title composite generated from the authoritative ICRS apparatus vectors, the fixed anisotropic G_phase matrix, and the exact 2:1 kinematic path. Force values are shown per unit kappa_F.

P. Dwayne Esterline, Principal Investigator
BS, Huntington University (1999)
MBA, Indiana Wesleyan University (2008)
Krytronx

Archival computational edition - Unix epoch 1784803951

Frontispiece: authoritative base cell, planes, and all nodes

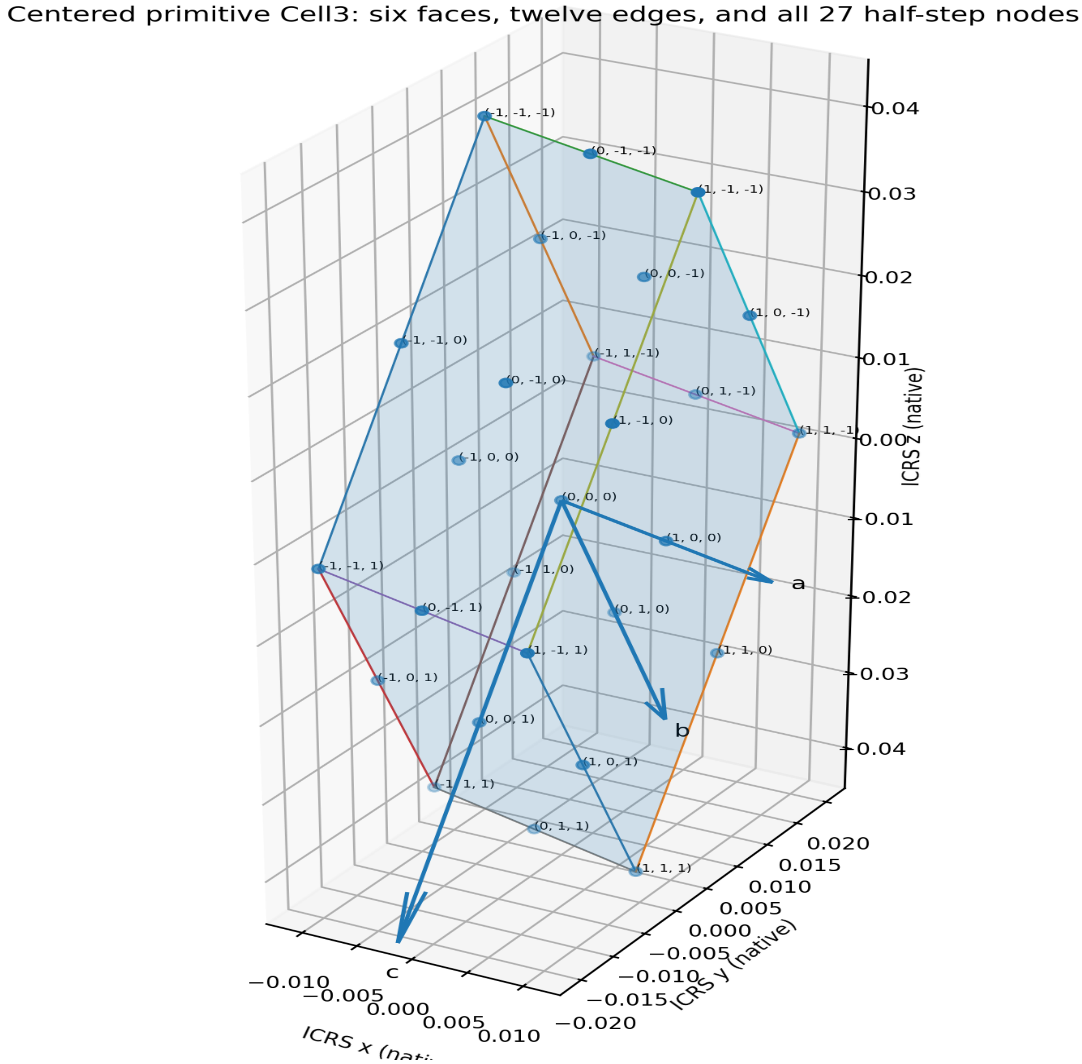

Figure 2. Dimensionally correct centered primitive Cell3 in ICRS native coordinates. All six boundary faces, twelve edges, twenty-seven centered half-step nodes, integer node addresses, and basis vectors are generated directly from the source basis matrix.

Abstract

This reference defines the complete computational foundation for the lateral force generated by the exact 2:1 dual-axis kinetic-mass path during translation through the anisotropic K-Field. Two equal endpoint masses remain antipodal on a rigid massless bar. The outer axis is locked to the Earth-CMB traversal direction, while the inner rotation proceeds at twice the outer angular rate. The endpoint velocities are the sum of the common Earth-CMB translation and equal opposite rotational velocities.

The K-Field traversal cost is evaluated with the fixed symmetric positive-definite matrix G_phase. The translation-rotation cross scalar B(phi)=u_phi^T G_phase V_E changes sign with rotor phase. Radial differentiation of the declared traversal cost gives two endpoint forces. Their pure-rotation terms cancel in the two-mass sum, while their translation-rotation terms reinforce. Orthogonal projection removes the CMB-axis component and yields the instantaneous lateral force.

At the declared normalization m=1 kg, R=0.5 m, omega_B=1 rad/s, omega_A=2 rad/s, and kappa_F=1, the exact outer and inner speeds are 9.5493 RPM and 19.0986 RPM. The continuous maximum lateral force is 4.133571e+06 N per unit kappa_F. The exact lateral impulse per outer cycle is pi kappa_F m c_hat cross(G_phase V_E), with magnitude 1.534054e+06 N s per unit kappa_F. The absolute engineering magnitude remains proportional to the constitutive gain kappa_F; the geometry, phase, vector direction, and scaling law are fixed independently of that calibration.

Central result. The complete code-stack kernel reduces to one precomputed anisotropic vector G_phase V_E, three local scalar projections, the exact 2:1 path, and the lateral projector I-c_hat c_hat^T. The cycle residue is nonzero exactly when G_phase V_E is not parallel to the CMB axis.

Keywords: K-Field, Cell3, Earth-CMB traversal, anisotropic G, compound rotation, kinetic-mass path, lateral force, transverse impulse, constitutive gain, code kernel.

Contents

1. Purpose and model lock

The objective is to define a compact but complete formula set suitable as the foundation of a production code stack for computation of the lateral force. The formulas must preserve the exact ICRS orientation, the declared 2:1 rate lock, the Earth-CMB translation state, the anisotropic G_phase metric, and the distinction between total, axial, and lateral force.

The computation uses a fixed K-Field metric orientation. The device translates through that structure with velocity V_E. The selected seasonal RA=0 degree datum may translate the absolute Earth-center path, but it does not rotate the apparatus frame and does not enter the local force equations. All values in this report use the declared native mechanical normalization unless an independently measured kappa_F is supplied.

Model lock. Do not replace the declared K-Field constitutive force with a standard centripetal-force formula. The rotor speed determines phase progression; the large translation-rotation scalar is generated by V_E entering the anisotropic traversal cost. Do not silently set kappa_F to a measured physical value; kappa_F=1 is a computational normalization.

2. Authoritative constants and coordinate frame

Quantity	Definition	Value
m	mass at each endpoint	1 kg
R	endpoint radius	0.5 m
omega_B	outer angular speed	1 rad/s = 9.549296586 RPM
omega_A	inner angular speed	2 rad/s = 19.098593171 RPM
kappa_F	constitutive traversal-force gain	1 (native normalization)
c_hat	Earth-CMB unit axis	(-0.970770855836, 0.207348009111, -0.120874929482)
e0	bar direction at phase zero	(-0.200864884110, -0.426231479950, 0.882031758970)
A0	inner-axis reference	(0.131366829113, 0.880530254232, 0.455422032396)
V_E	Earth-CMB velocity	(-359003.199066, 76679.886034, -44701.060101) m/s
|V_E|	Earth-CMB speed	369812.502000406 m/s

The ordered frame (e0,A0,c_hat) is orthonormal and right-handed. The source relation is A0=c_hat cross e0. Every vector operation must remain in this same ICRS frame.

```
G_phase =
[[ 2.9764106378,  1.2431507795, -2.0803600354],
 [ 1.2431507795,  2.5922286635, -2.3178839937],
 [-2.0803600354, -2.3178839937,  6.7735709133]]
```

The eigenvalues are 1.429328753, 2.112545026, and 8.800336435. All are positive, so G_phase is positive definite.

```
G_phase V_E = (-880221.981635091, -143911.436561599, 266335.026948619) m/s
```

The angle between c_hat and the normalized vector G_phase V_E is 31.640861271 degrees. This nonparallelism is the geometric source of the closed-cycle lateral residue.

3. Cell3 base geometry and plane system

The centered primitive Cell3 is generated by the three ICRS basis vectors a, b, and c. Its twenty-seven half-step nodes consist of eight corners, twelve edge centers, six face centers, and one body center. The six boundary faces form three opposite plane pairs.

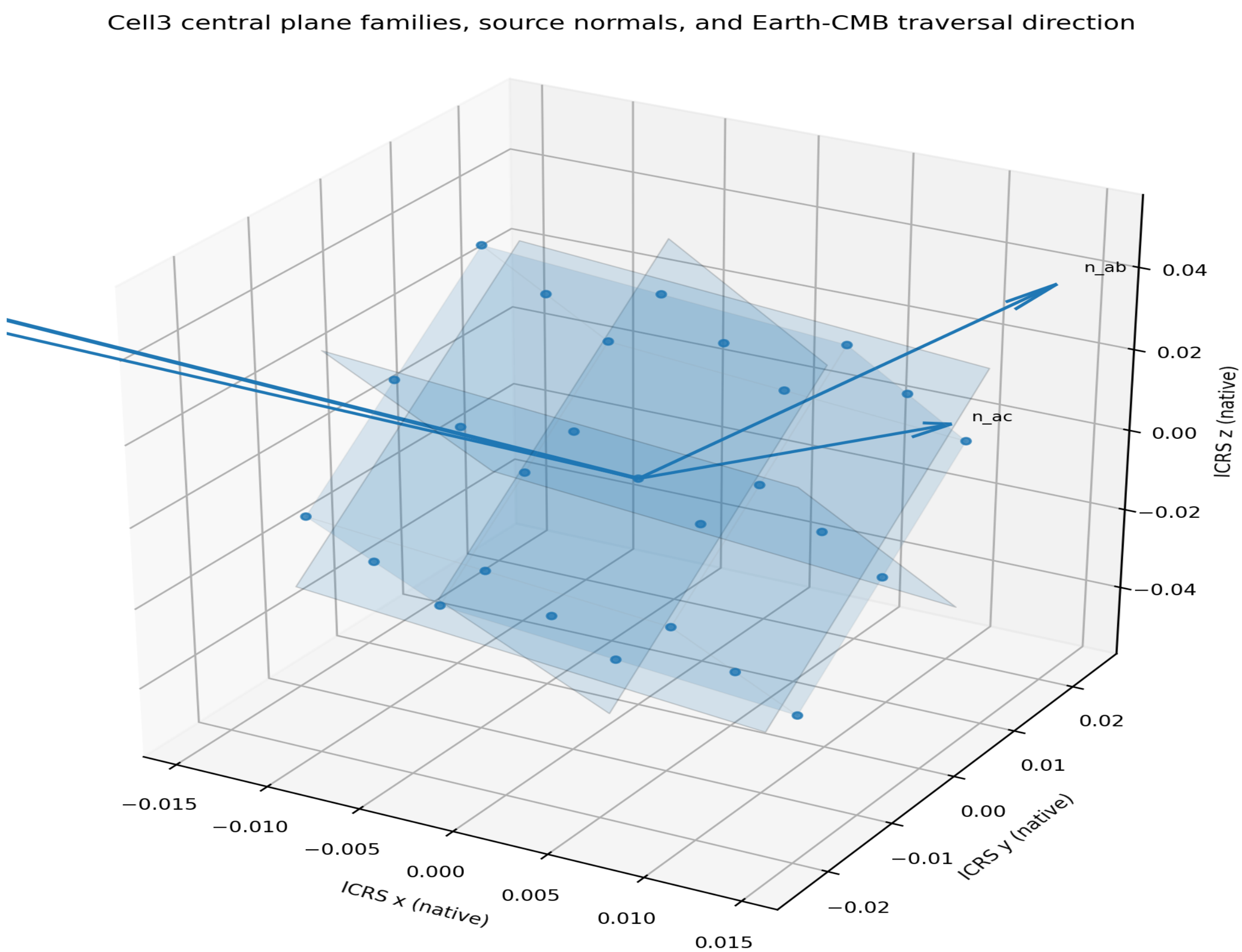

Figure 3. Central ab, ac, and bc plane families, source face normals, all nodes, and the Earth-CMB orientation vector. The cell and plane patches retain native geometric dimensions.

Basis	ICRS x	ICRS y	ICRS z	Length
a	0.020381438517	-0.003416414496	-0.002332965917	0.020797058782
b	0.002198745968	0.012902170259	-0.034066501536	0.036494205131
c	-0.000630379485	-0.024257722017	-0.042924354571	0.049308565900

Primitive Cell3 volume: 2.842066937809649e-05 native length cubed.

The Cell3 geometry is retained because it fixes the K-Field orientation, plane families, and node registry. The local lateral-force kernel itself uses G_phase, V_E, and the apparatus frame; the Cell3 tables provide the authoritative structural embedding and reproducibility lock.

4. Exact 2:1 compound-rotation kinematics

Let phi=omega_B t be the outer mechanical phase. The transverse basis rotates once about c_hat while the inner bar phase advances at 2phi.

```
e(phi) = cos(phi)e0 + sin(phi)A0
f(phi) = -sin(phi)e0 + cos(phi)A0
```

```
u(phi) = cos(2phi)e(phi) + sin(2phi)c_hat
```

The positive endpoint is r_plus=r_C+R u and the negative endpoint is r_minus=r_C-R u. Because e(phi) is perpendicular to c_hat, u has unit length at every phase.

```
u_phi = du/dphi
      = -2 sin(2phi)e(phi) + cos(2phi)f(phi) + 2 cos(2phi)c_hat
```

The fixed-frame harmonic expansion is useful for symbolic integration and an optimized code implementation:

```
u(phi) = 0.5[cos(3phi)+cos(phi)]e0
       + 0.5[sin(3phi)-sin(phi)]A0
       + sin(2phi)c_hat
```

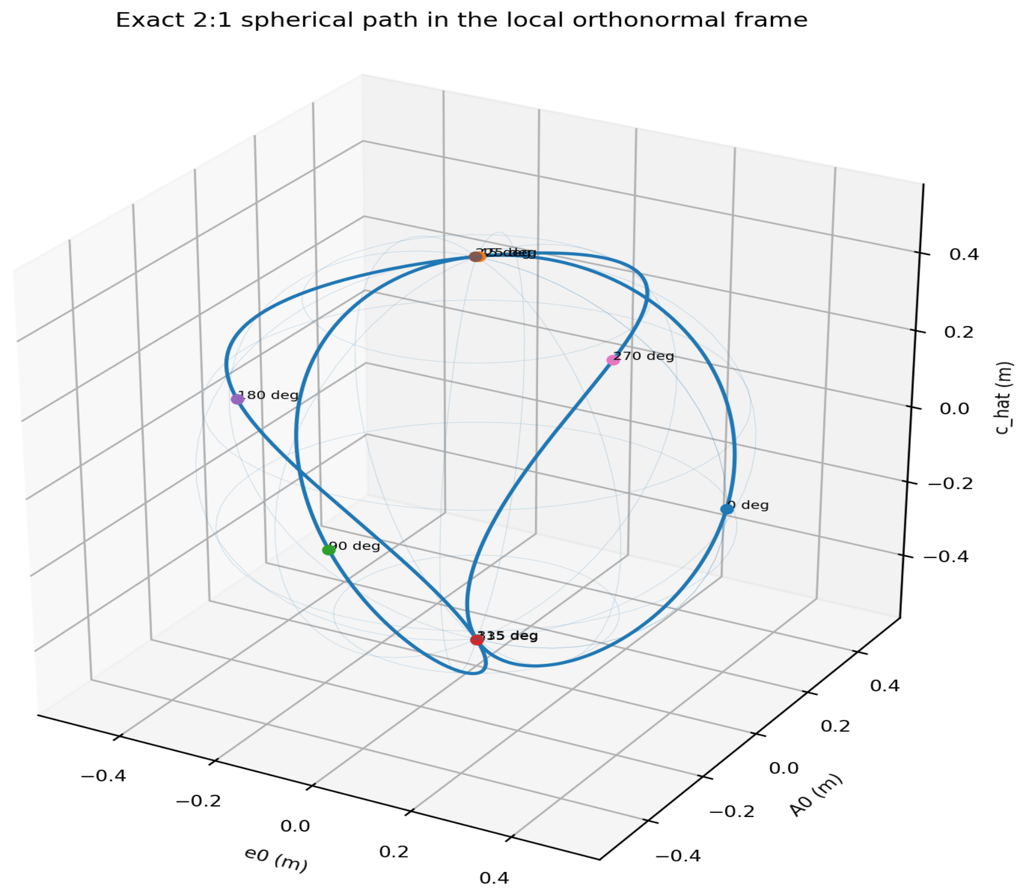

Figure 4. The exact local trajectory on the radius-0.5 m sphere. The path reaches equatorial states at multiples of 90 degrees and CMB-axis states at odd multiples of 45 degrees.

5. Anisotropic traversal cost and endpoint forces

The endpoint velocities in the K-Field frame are the common Earth-CMB translation plus equal opposite rotational velocities:

```
v_plus  = V_E + R omega_B u_phi
v_minus = V_E - R omega_B u_phi
```

The declared anisotropic traversal costs are

```
E_plus  = 0.5 m kappa_F v_plus^T  G_phase v_plus
E_minus = 0.5 m kappa_F v_minus^T G_phase v_minus
```

Define the two phase scalars

```
B(phi) = u_phi^T G_phase V_E
C(phi) = u_phi^T G_phase u_phi
```

The endpoint cost difference isolates the advancing-retreating translation channel:

```
E_plus - E_minus = 2 m kappa_F R omega_B B(phi)
```

Resolving the radial derivatives on each endpoint local outward direction gives

```
F_plus  = -kappa_F m [omega_B B + R omega_B^2 C] u
F_minus =  kappa_F m [R omega_B^2 C - omega_B B] u
```

Adding the endpoint vectors cancels the equal pure-rotation terms and reinforces the translation-rotation terms:

```
F_net(phi) = F_plus + F_minus
           = -2 kappa_F m omega_B B(phi) u(phi)
```

The force magnitude is therefore

```
|F_net| = 2 kappa_F m omega_B |B(phi)|
```

At the declared slow rotation rate, the meganeuton-scale normalized values do not arise from conventional centripetal acceleration. They arise because the model multiplies the approximately 369.8 km/s traversal velocity by the anisotropic metric and the phase derivative. The engineering magnitude remains proportional to kappa_F.

6. Lateral-force projection

The lateral direction is the plane perpendicular to the Earth-CMB axis. Its projection matrix is

```
P_perp = I - c_hat c_hat^T
```

The lateral-force vector is

```
F_lateral = P_perp F_net
```

Since u dot c_hat=sin(2phi), the endpoint radial direction decomposes as

```
u = cos(2phi)e(phi) + sin(2phi)c_hat
```

Substitution gives the explicit lateral kernel:

```
F_lateral(phi)
 = -2 kappa_F m omega_B B(phi) cos(2phi)e(phi)
```

The signed scalar along the rotating transverse direction e(phi) is

```
F_lateral_signed
 = -2 kappa_F m omega_B B(phi) cos(2phi)
```

and the magnitude is

```
|F_lateral|
 = 2 kappa_F m omega_B |B(phi) cos(2phi)|
```

The corresponding axial channel is

```
F_parallel_scalar
 = F_net dot c_hat
 = -2 kappa_F m omega_B B(phi) sin(2phi)
```

The decomposition must satisfy F_net=F_lateral+F_parallel_scalar c_hat at every phase.

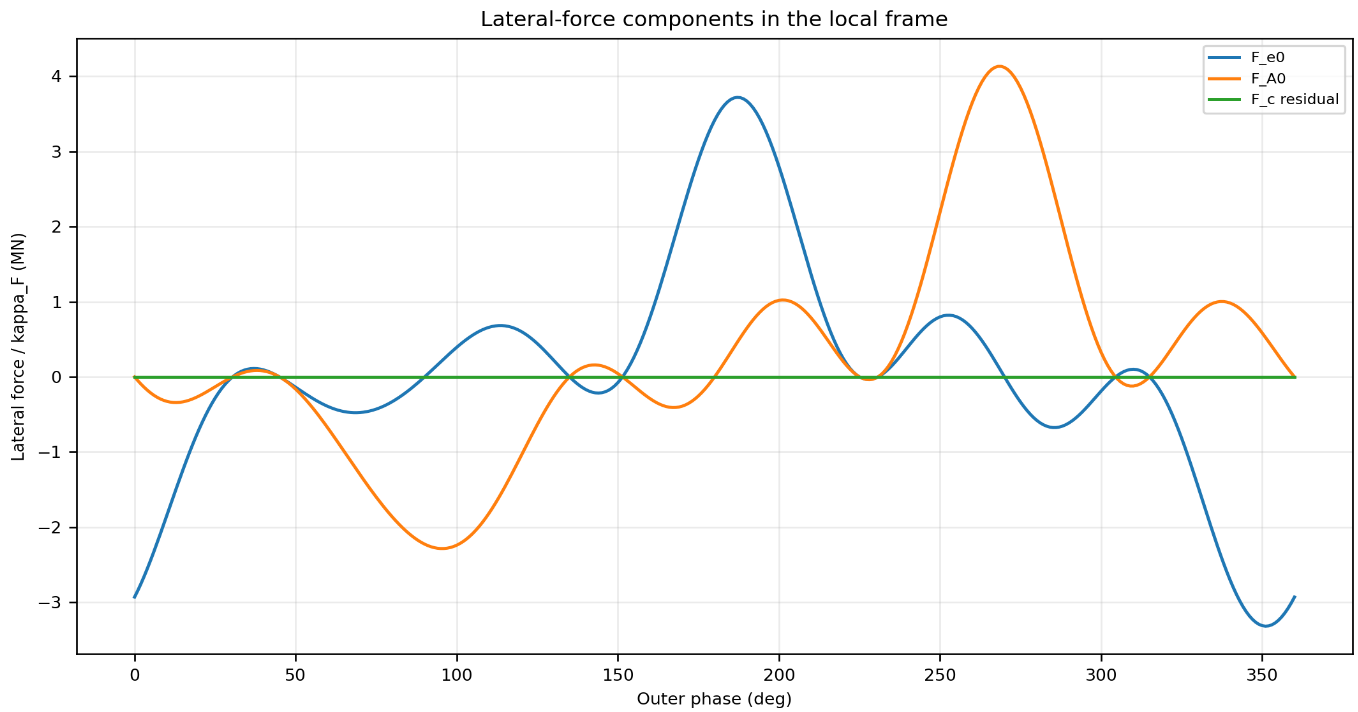

Figure 5. Lateral-force components in the local frame. The computed c_hat component is numerical zero; the two transverse components rotate with e(phi).

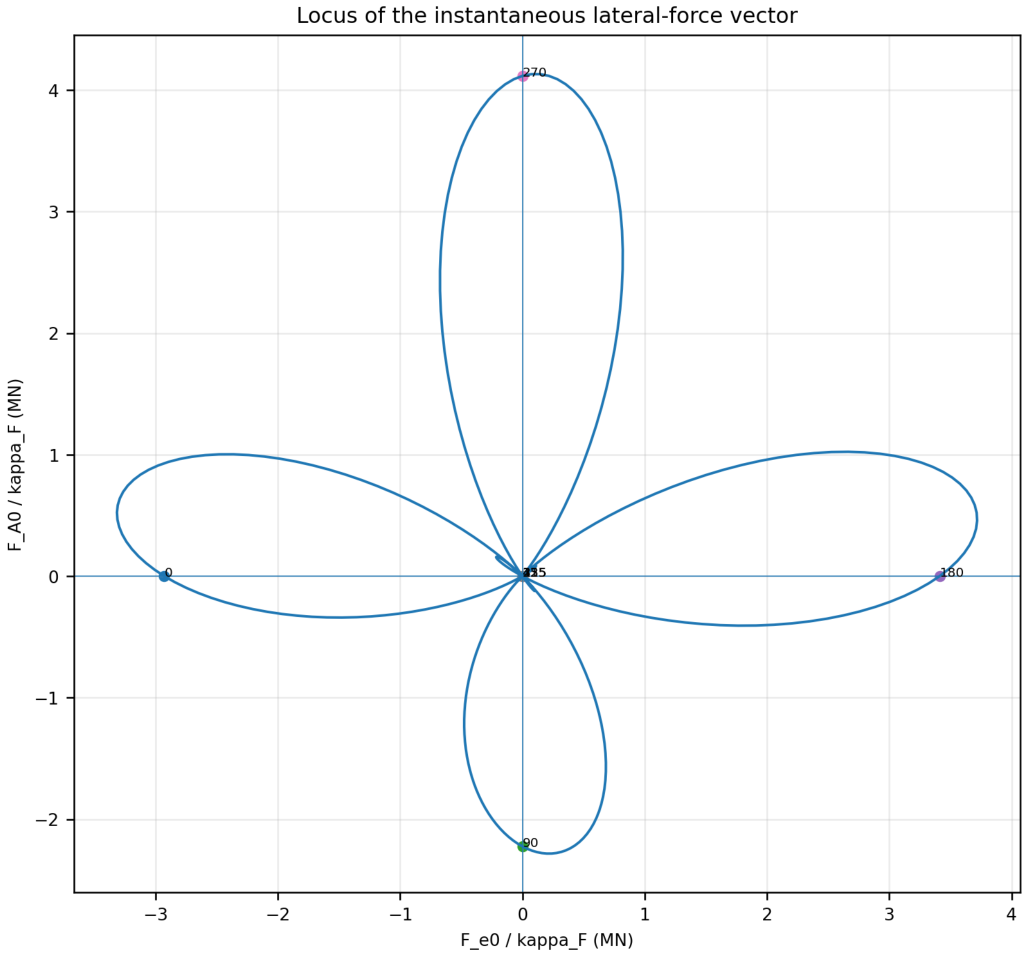

Figure 6. Locus of the instantaneous lateral-force vector in the fixed transverse (e0,A0) plane. Phase labels are in degrees.

7. Optimized harmonic computation

A high-throughput implementation should precompute G_phase V_E and its three projections on the local orthonormal frame:

```
p0 = e0^T G_phase V_E
p1 = A0^T G_phase V_E
p2 = c_hat^T G_phase V_E
```

For the authoritative inputs:

```
p0 = 473061.223213950288 m/s
p1 = -121055.505194313228 m/s
p2 = 792460.868977151928 m/s
```

Then B(phi) is evaluated without a matrix multiplication:

```
B(phi) = -0.5[3 sin(3phi)+sin(phi)]p0
       + 0.5[3 cos(3phi)-cos(phi)]p1
       + 2 cos(2phi)p2
```

Numerically:

```
B(phi) =
 -236530.611606975144 sin(phi)
 -709591.834820925491 sin(3phi)
 + 60527.752597156614 cos(phi)
 -181583.257791469834 cos(3phi)
 + 1584921.737954303855 cos(2phi)  m/s
```

The optimized lateral-only loop requires B(phi), e(phi), cos(2phi), and one scalar-vector multiplication.

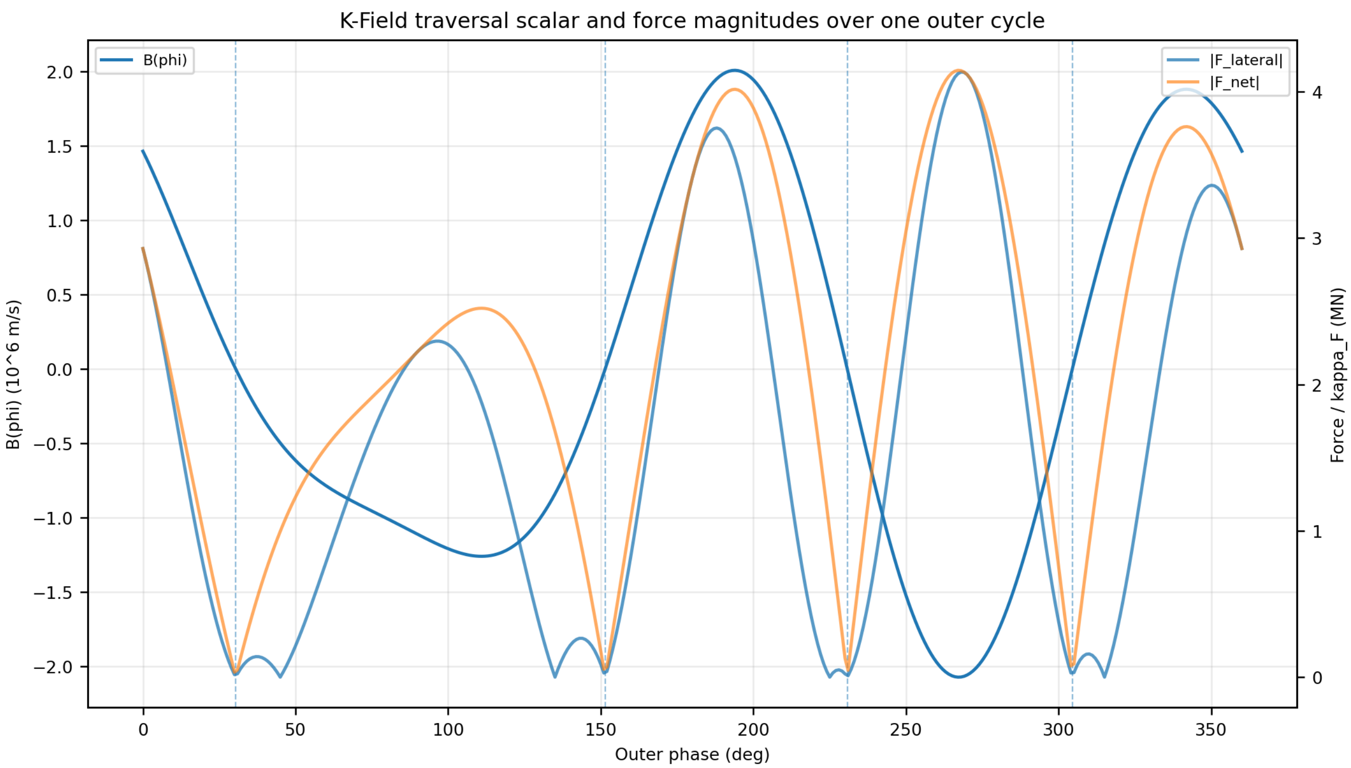

Figure 7. The traversal scalar B(phi), total force magnitude, and lateral-force magnitude. Dashed vertical lines mark the four roots of B(phi).

8. Numerical force structure

Quantity	Value per kappa_F	Phase
continuous maximum total force	4.145150737450e+06 N	267.242565394 deg
continuous maximum lateral force	4.133570511912e+06 N	268.335926872 deg
continuous maximum absolute axial force	2.897630308442e+06 N	330.256507604 deg
continuous RMS lateral force	2.045785361809e+06 N	full cycle
continuous RMS axial force	1.425104510386e+06 N	full cycle

The complete force vanishes where B(phi)=0. The four roots are:

```
root 1: 30.402363199422 deg
root 2: 151.435410918357 deg
root 3: 230.733949264391 deg
root 4: 304.459020415836 deg
```

The geometric phase classes are independent of kappa_F:

Condition	Phase class	Force direction
sin(2phi)=0	0, 90, 180, 270 deg	purely lateral when B is nonzero
cos(2phi)=0	45, 135, 225, 315 deg	purely axial when B is nonzero
B(phi)=0	four metric roots	complete force equals zero
otherwise	mixed phase	simultaneous lateral and axial components

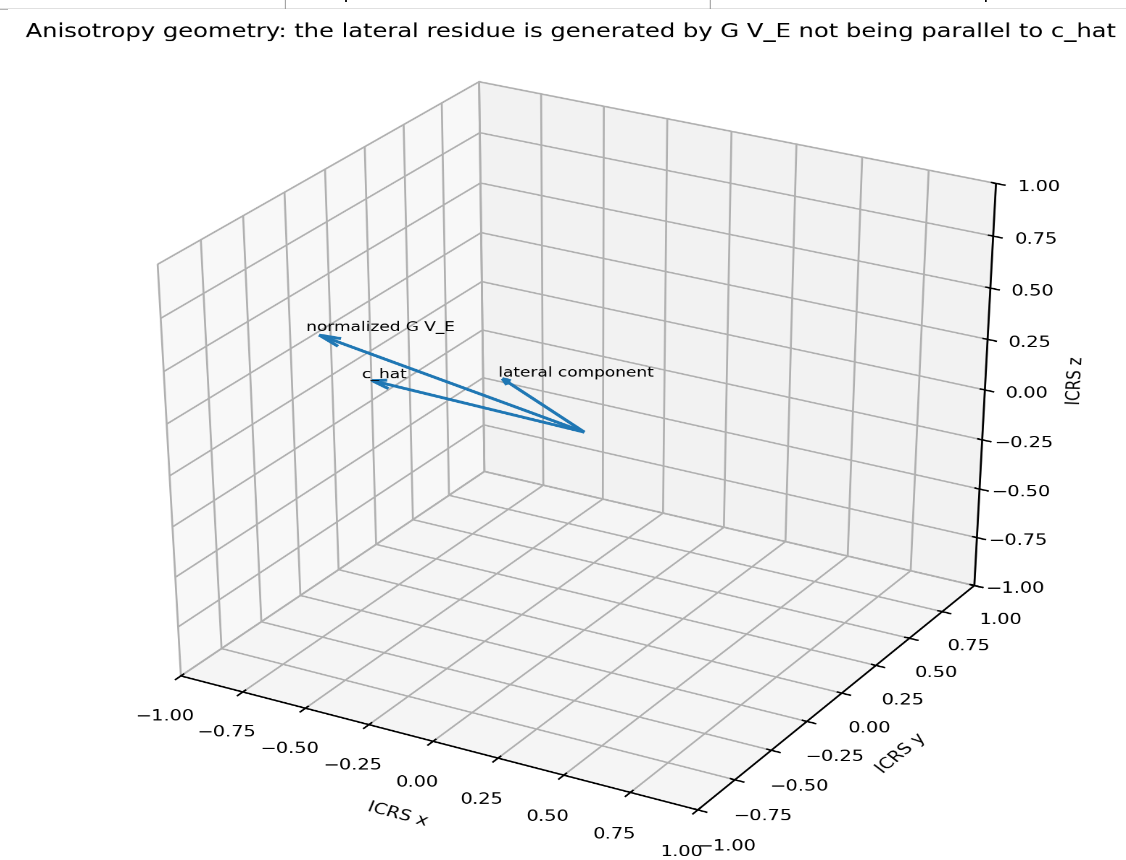

Figure 8. The anisotropy geometry. The normalized vector G_phase V_E is not parallel to c_hat. Its transverse part generates the exact cycle residue.

9. Exact cycle impulse and mean lateral force

The outer period is T_B=2pi/omega_B. With dphi=omega_B dt, the explicit omega_B factor cancels from the cycle integral.

```
J = integral_0^T_B F_net(t) dt
  = pi kappa_F m c_hat cross(G_phase V_E)
```

The complete cycle impulse is already lateral because a cross product with c_hat is perpendicular to c_hat:

```
J_lateral = J
c_hat dot J = 0
```

Numerically, per unit kappa_F:

```
J = (118842.532042625317, 1146514.977539591957, 1012275.514801947284) N s
|J| = 1534054.060003252001 N s
```

The cycle-average lateral force is

```
 = J/T_B
             = (kappa_F m omega_B/2)c_hat cross(G_phase V_E)
```

```
 = (18914.376424140781, 182473.526004319428, 161108.651951622975) N
|| = 244152.286619708553 N
```

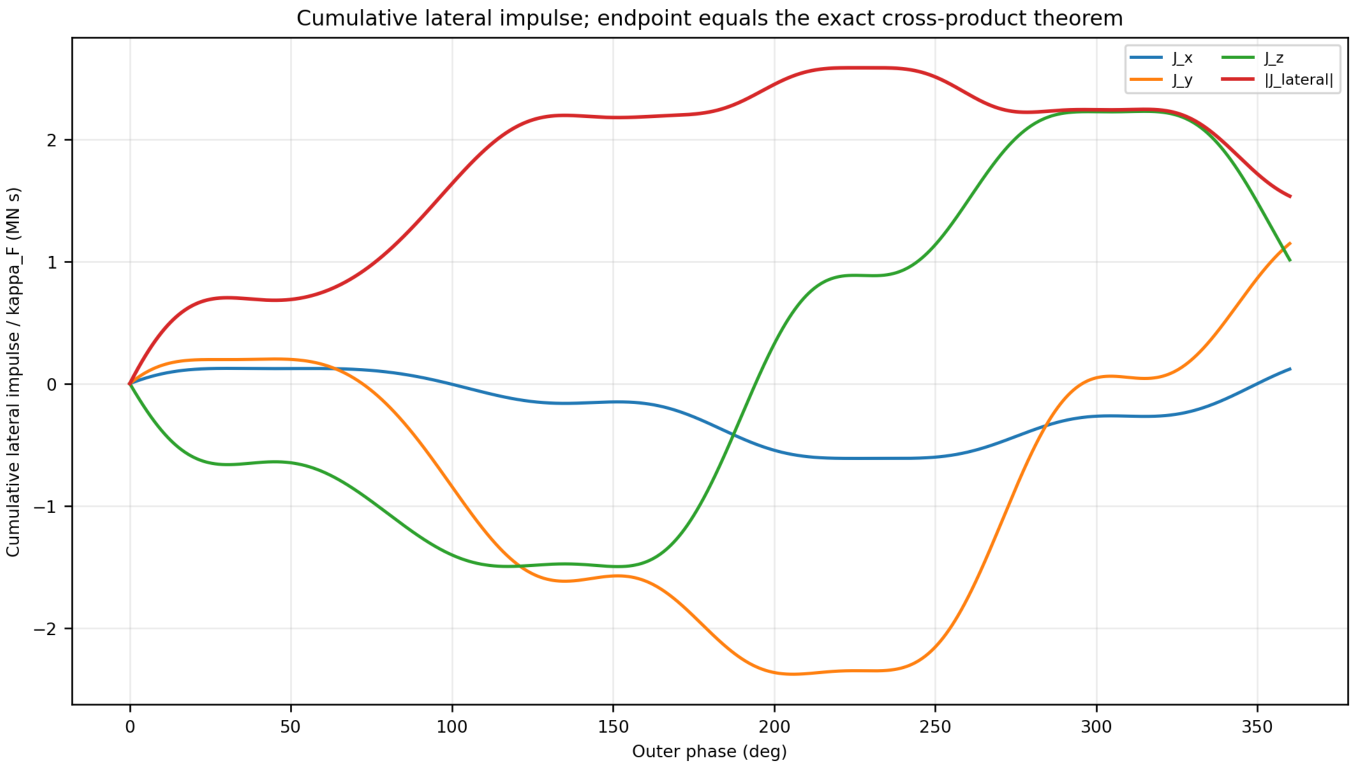

Figure 9. Cumulative lateral impulse over one outer rotation. The endpoint at 360 degrees reproduces the exact cross-product vector.

10. Anisotropy condition and directional interpretation

The exact cycle magnitude may be written

```
|J| = pi kappa_F m sqrt[V_E^T G_phase^2 V_E
      - (c_hat^T G_phase V_E)^2]
```

when V_E is parallel to c_hat. The residue vanishes exactly when

```
G_phase V_E is parallel to c_hat
```

Therefore:

```
J != 0  iff  c_hat cross(G_phase V_E) != 0
```

An isotropic metric G_phase=lambda I gives J=0. Traversal along any principal eigenaxis of G_phase also gives J=0. The authoritative Earth-CMB direction is not an eigenaxis, so the computed residue is nonzero.

The dimensionless directional cost along the CMB axis is f_G(c_hat)=sqrt(c_hat^T G_phase c_hat)=1.463855196277. The anisotropy residue norm ||c_hat cross(G_phase c_hat)|| is 1.320411209995.

11. Scaling, RPM, and engineering normalization

The declared angular speeds correspond to

```
RPM_B = omega_B 60/(2pi) = 9.549296585514 RPM
RPM_A = omega_A 60/(2pi) = 19.098593171027 RPM
```

The instantaneous lateral force scales as

```
F_lateral proportional to kappa_F m omega_B
```

The impulse per outer cycle scales as

```
J_lateral proportional to kappa_F m
```

and is independent of R and omega_B for the declared radial force law. The cycle-average lateral force scales linearly with omega_B because the same impulse is delivered in a shorter period.

Magnitude interpretation. The numerical values printed at kappa_F=1 are native normalization coefficients, not calibrated engineering predictions. A measured or independently derived kappa_F must multiply every force and impulse. For example, kappa_F=10^-6 reduces a 4.13 MN normalized peak lateral coefficient to approximately 4.13 N.

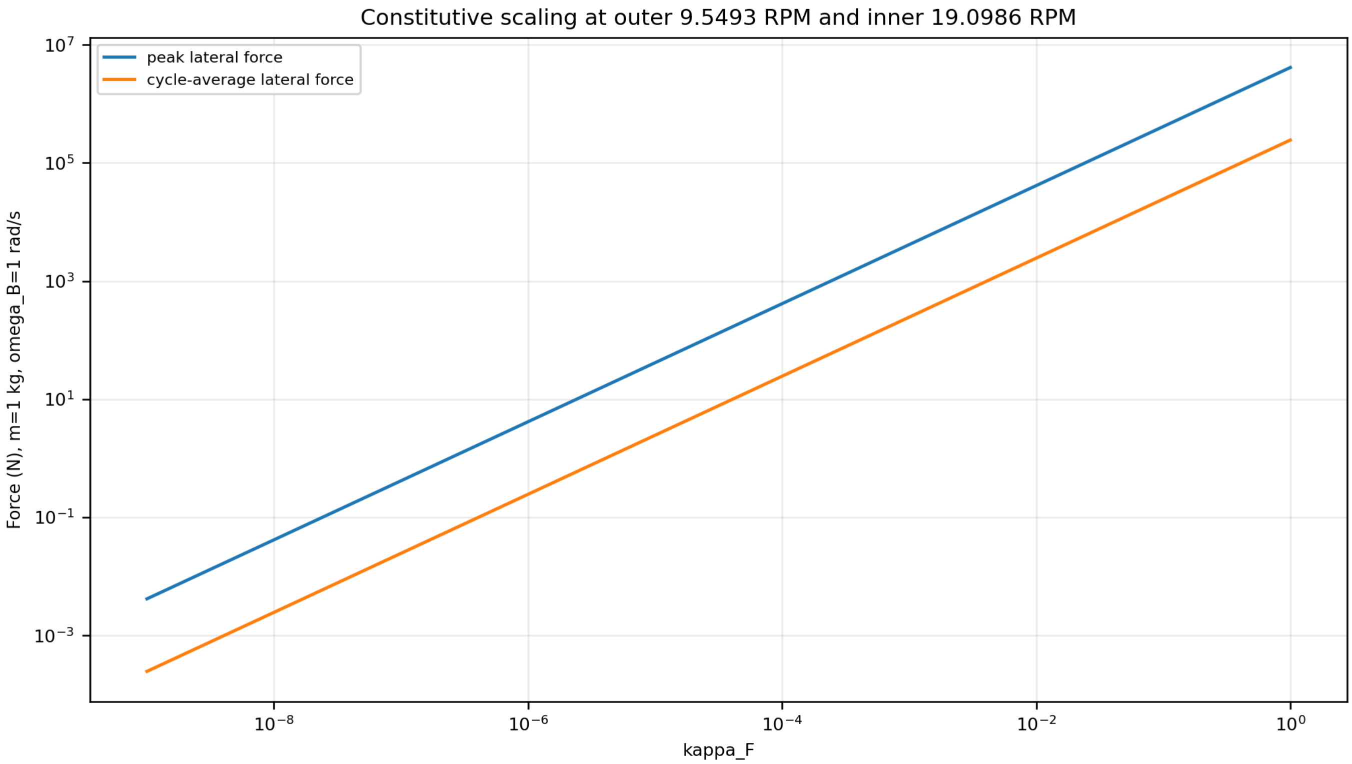

Figure 10. Linear constitutive scaling shown on logarithmic axes at the declared outer and inner RPM.

12. Code-stack architecture

The recommended implementation separates immutable geometry, operating state, phase evaluation, vector output, cycle integration, and validation.

12.1 Immutable geometry

```
store G_phase, c_hat, e0, A0
validate symmetry, positive definiteness, unit norms, orthogonality, handedness
precompute GV=G_phase@V_E and p0,p1,p2
```

12.2 Operating state

```
store m, R, omega_B, kappa_F
set omega_A=2 omega_B
phi=omega_B t
```

12.3 Phase kernel

```
e = cos(phi)e0 + sin(phi)A0
B = -0.5(3sin3phi+sinphi)p0
    +0.5(3cos3phi-cosphi)p1
    +2cos2phi p2
F_lat_signed = -2 kappa_F m omega_B B cos2phi
F_lateral = F_lat_signed e
```

12.4 Optional full-force kernel

```
u = cos2phi e + sin2phi c_hat
F_net = -2 kappa_F m omega_B B u
F_parallel_scalar = F_net dot c_hat
F_lateral = F_net - F_parallel_scalar c_hat
```

12.5 Cycle kernel

```
J_exact = pi kappa_F m cross(c_hat,GV)
F_average = omega_B J_exact/(2pi)
```

12.6 Recommended outputs

For each phase, emit phase, e, u, u_phi, B, C, endpoint velocities, endpoint costs, endpoint forces, total force, signed lateral scalar, lateral vector, lateral magnitude, axial scalar, and validation residuals. At cycle closure, emit the numerical impulse, analytic impulse, difference norm, mean force, and orthogonality residual.

13. Validation gates and numerical tolerances

A production implementation should reject inputs that violate the frame or metric locks.

```
||G_phase-G_phase^T|| < epsilon_G
lambda_min(G_phase) > 0
||c_hat|| = ||e0|| = ||A0|| = 1
e0 dot c_hat = A0 dot c_hat = e0 dot A0 = 0
det[e0,A0,c_hat] > 0
```

Every evaluated phase should satisfy

```
||u|| = 1
F_lateral dot c_hat = 0
F_net = F_lateral + F_parallel_scalar c_hat
|F_net|^2 = |F_lateral|^2 + F_parallel_scalar^2
B_direct = B_harmonic
```

Cycle closure should satisfy

```
||J_numeric - pi kappa_F m cross(c_hat,G_phase V_E)|| < epsilon_J
|c_hat dot J_numeric| < epsilon_orth
```

The report-generation audit repeated the reconstruction and closure gates ten times. The analytic and dense numerical cycle impulses agree within 5.393e-09 N s.

14. Canonical formula ledger

ID	Canonical expression	Purpose
K01	phi=omega_B t	outer phase
K02	A0=c_hat cross e0	right-handed frame
K03	e=cos(phi)e0+sin(phi)A0	rotating transverse direction
K04	f=-sin(phi)e0+cos(phi)A0	phase derivative of e
K05	u=cos(2phi)e+sin(2phi)c_hat	endpoint radial direction
K06	u_phi=-2sin(2phi)e+cos(2phi)f+2cos(2phi)c_hat	phase tangent
K07	v_plus/minus=V_E +/- R omega_B u_phi	endpoint velocities
K08	B=u_phi^T G_phase V_E	translation-rotation scalar
K09	C=u_phi^T G_phase u_phi	pure-rotation scalar
K10	F_net=-2 kappa_F m omega_B B u	two-mass force
K11	P_perp=I-c_hat c_hat^T	lateral projector
K12	F_lateral=P_perp F_net	lateral vector
K13	F_lateral=-2 kappa_F m omega_B B cos(2phi)e	optimized lateral vector
K14	F_parallel=-2 kappa_F m omega_B B sin(2phi)	axial scalar
K15	J=pi kappa_F m c_hat cross(G_phase V_E)	cycle lateral impulse
K16	=omega_B J/(2pi)	cycle-average lateral force
K17	RPM=omega 60/(2pi)	angular-speed conversion

Appendix A. Cell3 boundary vertices and plane families

#	i	j	k	ICRS x	ICRS y	ICRS z
0	-0.5	-0.5	-0.5	-0.010974902500	0.007385983127	0.039661911012
1	-0.5	-0.5	0.5	-0.011605281985	-0.016871738890	-0.003262443559
2	-0.5	0.5	-0.5	-0.008776156532	0.020288153386	0.005595409476
3	-0.5	0.5	0.5	-0.009406536017	-0.003969568631	-0.037328945095
4	0.5	-0.5	-0.5	0.009406536017	0.003969568631	0.037328945095
5	0.5	-0.5	0.5	0.008776156532	-0.020288153386	-0.005595409476
6	0.5	0.5	-0.5	0.011605281985	0.016871738890	0.003262443559
7	0.5	0.5	0.5	0.010974902500	-0.007385983127	-0.039661911012

Family	Area native^2	Included angle	Unit normal ICRS	Boundary |d|
A_ab	7.547217896726e-04	83.933483012 deg	(0.1940922008, 0.9131771688, 0.3583792348)	0.018828573500
A_ac	1.011255757688e-03	80.448130018 deg	(0.0890525941, 0.8665767693, -0.4910339483)	0.014052166903
A_bc	1.385784431733e-03	50.363231814 deg	(-0.9959651903, 0.0836021625, -0.0326192906)	0.010254361619

Appendix B. Complete 27-node centered half-step registry

#	2i	2j	2k	lambda	ICRS x	ICRS y	ICRS z
0	-1	-1	-1	-3	-0.010974902500	0.007385983127	0.039661911012
1	-1	-1	0	-2	-0.011290092243	-0.004742877881	0.018199733726
2	-1	-1	1	-1	-0.011605281985	-0.016871738890	-0.003262443559
3	-1	0	-1	-2	-0.009875529516	0.013837068257	0.022628660244
4	-1	0	0	-1	-0.010190719258	0.001708207248	0.001166482958
5	-1	0	1	0	-0.010505909001	-0.010420653760	-0.020295694327
6	-1	1	-1	-1	-0.008776156532	0.020288153386	0.005595409476
7	-1	1	0	0	-0.009091346274	0.008159292378	-0.015866767810
8	-1	1	1	1	-0.009406536017	-0.003969568631	-0.037328945095
9	0	-1	-1	-2	-0.000784183241	0.005677775879	0.038495428054
10	0	-1	0	-1	-0.001099372984	-0.006451085129	0.017033250768
11	0	-1	1	0	-0.001414562727	-0.018579946138	-0.004428926518
12	0	0	-1	-1	0.000315189743	0.012128861009	0.021462177286
13	0	0	0	0	0.000000000000	0.000000000000	0.000000000000
14	0	0	1	1	-0.000315189743	-0.012128861009	-0.021462177286
15	0	1	-1	0	0.001414562727	0.018579946138	0.004428926518
16	0	1	0	1	0.001099372984	0.006451085129	-0.017033250768
17	0	1	1	2	0.000784183241	-0.005677775879	-0.038495428054
18	1	-1	-1	-1	0.009406536017	0.003969568631	0.037328945095
19	1	-1	0	0	0.009091346274	-0.008159292378	0.015866767810
20	1	-1	1	1	0.008776156532	-0.020288153386	-0.005595409476
21	1	0	-1	0	0.010505909001	0.010420653760	0.020295694327
22	1	0	0	1	0.010190719258	-0.001708207248	-0.001166482958
23	1	0	1	2	0.009875529516	-0.013837068257	-0.022628660244
24	1	1	-1	1	0.011605281985	0.016871738890	0.003262443559
25	1	1	0	2	0.011290092243	0.004742877881	-0.018199733726
26	1	1	1	3	0.010974902500	-0.007385983127	-0.039661911012

Appendix C. Complete one-degree lateral-force ledger

All values use m=1 kg, R=0.5 m, omega_B=1 rad/s, omega_A=2 rad/s, and kappa_F=1. Multiply force columns by the desired kappa_F. The repeated 360-degree row is the closure endpoint.

deg	B (m/s)	F_lat signed (N)	|F_lat| (N)	F_lat x	F_lat y	F_lat z	F_parallel (N)
0	1.463866e+06	-2.927732e+06	2.927732e+06	5.880786e+05	1.247892e+06	-2.582353e+06	-9.940019e-11
1	1.421875e+06	-2.842018e+06	2.842018e+06	5.642589e+05	1.167499e+06	-2.528957e+06	-9.924546e+04
2	1.378536e+06	-2.750356e+06	2.750356e+06	5.395040e+05	1.087056e+06	-2.468138e+06	-1.923236e+05
3	1.333953e+06	-2.653291e+06	2.653291e+06	5.139806e+05	1.007093e+06	-2.400320e+06	-2.788721e+05
4	1.288230e+06	-2.551387e+06	2.551387e+06	4.878556e+05	9.281194e+05	-2.325977e+06	-3.585741e+05
5	1.241474e+06	-2.445226e+06	2.445226e+06	4.612947e+05	8.506117e+05	-2.245617e+06	-4.311593e+05
6	1.193788e+06	-2.335401e+06	2.335401e+06	4.344615e+05	7.750169e+05	-2.159789e+06	-4.964048e+05
7	1.145276e+06	-2.222513e+06	2.222513e+06	4.075157e+05	7.017468e+05	-2.069069e+06	-5.541347e+05
8	1.096043e+06	-2.107168e+06	2.107168e+06	3.806122e+05	6.311755e+05	-1.974059e+06	-6.042208e+05
9	1.046191e+06	-1.989974e+06	1.989974e+06	3.539002e+05	5.636375e+05	-1.875384e+06	-6.465818e+05
10	9.958227e+05	-1.871535e+06	1.871535e+06	3.275217e+05	4.994257e+05	-1.773681e+06	-6.811829e+05
11	9.450374e+05	-1.752447e+06	1.752447e+06	3.016109e+05	4.387905e+05	-1.669600e+06	-7.080345e+05
12	8.939341e+05	-1.633299e+06	1.633299e+06	2.762934e+05	3.819384e+05	-1.563794e+06	-7.271915e+05
13	8.426095e+05	-1.514665e+06	1.514665e+06	2.516852e+05	3.290321e+05	-1.456915e+06	-7.387514e+05
14	7.911584e+05	-1.397103e+06	1.397103e+06	2.278924e+05	2.801904e+05	-1.349613e+06	-7.428528e+05
15	7.396732e+05	-1.281152e+06	1.281152e+06	2.050103e+05	2.354885e+05	-1.242524e+06	-7.396732e+05
16	6.882438e+05	-1.167328e+06	1.167328e+06	1.831235e+05	1.949589e+05	-1.136271e+06	-7.294273e+05
17	6.369573e+05	-1.056123e+06	1.056123e+06	1.623051e+05	1.585928e+05	-1.031456e+06	-7.123640e+05
18	5.858981e+05	-9.480031e+05	9.480031e+05	1.426169e+05	1.263417e+05	-9.286593e+05	-6.887646e+05
19	5.351475e+05	-8.434040e+05	8.434040e+05	1.241091e+05	9.811924e+04	-8.284322e+05	-6.589394e+05
20	4.847836e+05	-7.427315e+05	7.427315e+05	1.068205e+05	7.380337e+04	-7.312951e+05	-6.232257e+05
21	4.348809e+05	-6.463590e+05	6.463590e+05	9.077844e+04	5.323900e+04	-6.377341e+05	-5.819842e+05
22	3.855108e+05	-5.546265e+05	5.546265e+05	7.599924e+04	3.624068e+04	-5.481982e+05	-5.355965e+05
23	3.367407e+05	-4.678395e+05	4.678395e+05	6.248838e+04	2.259551e+04	-4.630966e+05	-4.844620e+05
24	2.886346e+05	-3.862685e+05	3.862685e+05	5.024097e+04	1.206622e+04	-3.827970e+05	-4.289946e+05
25	2.412524e+05	-3.101481e+05	3.101481e+05	3.924223e+04	4.394409e+03	-3.076241e+05	-3.696201e+05
26	1.946502e+05	-2.396773e+05	2.396773e+05	2.946802e+04	-6.962004e+02	-2.378578e+05	-3.067729e+05
27	1.488801e+05	-1.750191e+05	1.750191e+05	2.088549e+04	-3.496416e+03	-1.737333e+05	-2.408931e+05
28	1.039902e+05	-1.163012e+05	1.163012e+05	1.345374e+04	-4.308243e+03	-1.154400e+05	-1.724236e+05
29	6.002431e+04	-6.361608e+04	6.361608e+04	7.124523e+03	-3.441572e+03	-6.312212e+04	-1.018070e+05
30	1.702230e+04	-1.702230e+04	1.702230e+04	1.843016e+03	-1.210930e+03	-1.687885e+04	-2.948349e+04
31	-2.498021e+04	2.345499e+04	2.345499e+04	-2.451419e+03	2.067673e+03	2.323471e+04	4.411243e+04
32	-6.595172e+04	5.782266e+04	5.782266e+04	-5.824435e+03	6.079772e+03	5.720640e+04	1.185540e+05
33	-1.058649e+05	8.611828e+04	8.611828e+04	-8.345892e+03	1.051533e+04	8.506546e+04	1.934248e+05
34	-1.446966e+05	1.084086e+05	1.084086e+05	-1.008906e+04	1.507144e+04	1.068807e+05	2.683207e+05
35	-1.824277e+05	1.247879e+05	1.247879e+05	-1.112983e+04	1.945481e+04	1.227585e+05	3.428519e+05
36	-2.190431e+05	1.353761e+05	1.353761e+05	-1.154590e+04	2.338408e+04	1.328404e+05	4.166447e+05
37	-2.545319e+05	1.403170e+05	1.403170e+05	-1.141609e+04	2.659185e+04	1.373004e+05	4.893434e+05
38	-2.888868e+05	1.397761e+05	1.397761e+05	-1.081954e+04	2.882652e+04	1.363427e+05	5.606113e+05
39	-3.221048e+05	1.339387e+05	1.339387e+05	-9.835052e+03	2.985381e+04	1.301983e+05	6.301320e+05
40	-3.541862e+05	1.230076e+05	1.230076e+05	-8.540471e+03	2.945806e+04	1.191224e+05	6.976105e+05
41	-3.851352e+05	1.072009e+05	1.072009e+05	-7.012039e+03	2.744331e+04	1.033912e+05	7.627741e+05
42	-4.149594e+05	8.675015e+04	8.675015e+04	-5.323868e+03	2.363404e+04	8.329872e+04	8.253725e+05
43	-4.436701e+05	6.189773e+04	6.189773e+04	-3.547440e+03	1.787571e+04	5.915406e+04	8.851787e+05
44	-4.712816e+05	3.289498e+04	3.289498e+04	-1.751165e+03	1.003502e+04	3.127797e+04	9.419889e+05
45	-4.978113e+05	6.096430e-11	1.372825e-10	1.164153e-10	5.820766e-11	4.365575e-11	9.956225e+05
46	-5.232797e+05	-3.652440e+04	3.652440e+04	1.644879e+03	-1.232023e+04	-3.434440e+04	1.045922e+06
47	-5.477102e+05	-7.641266e+04	7.641266e+04	3.126340e+03	-2.699582e+04	-7.141671e+04	1.092752e+06

deg	B (m/s)	F_lat signed (N)	|F_lat| (N)	F_lat x	F_lat y	F_lat z	F_parallel (N)
48	-5.711286e+05	-1.193984e+05	1.193984e+05	4.391504e+03	-4.407675e+04	-1.108780e+05	1.136000e+06
49	-5.935635e+05	-1.652161e+05	1.652161e+05	5.391898e+03	-6.359361e+04	-1.523915e+05	1.175574e+06
50	-6.150454e+05	-2.136030e+05	2.136030e+05	6.083549e+03	-8.555848e+04	-1.956246e+05	1.211403e+06
51	-6.356073e+05	-2.643004e+05	2.643004e+05	6.427036e+03	-1.099660e+05	-2.402517e+05	1.243436e+06
52	-6.552838e+05	-3.170550e+05	3.170550e+05	6.387484e+03	-1.367943e+05	-2.859552e+05	1.271638e+06
53	-6.741113e+05	-3.716205e+05	3.716205e+05	5.934516e+03	-1.660067e+05	-3.324280e+05	1.295995e+06
54	-6.921277e+05	-4.277585e+05	4.277585e+05	5.042168e+03	-1.975523e+05	-3.793744e+05	1.316505e+06
55	-7.093723e+05	-4.852392e+05	4.852392e+05	3.688768e+03	-2.313678e+05	-4.265120e+05	1.333184e+06
56	-7.258854e+05	-5.438429e+05	5.438429e+05	1.856778e+03	-2.673788e+05	-4.735718e+05	1.346058e+06
57	-7.417082e+05	-6.033598e+05	6.033598e+05	-4.673845e+02	-3.055007e+05	-5.203001e+05	1.355168e+06
58	-7.568825e+05	-6.635909e+05	6.635909e+05	-3.293546e+03	-3.456404e+05	-5.664580e+05	1.360563e+06
59	-7.714507e+05	-7.243483e+05	7.243483e+05	-6.627995e+03	-3.876975e+05	-6.118228e+05	1.362301e+06
60	-7.854553e+05	-7.854553e+05	7.854553e+05	-1.047371e+04	-4.315651e+05	-6.561873e+05	1.360448e+06
61	-7.989389e+05	-8.467462e+05	8.467462e+05	-1.483058e+04	-4.771313e+05	-6.993604e+05	1.355077e+06
62	-8.119440e+05	-9.080666e+05	9.080666e+05	-1.969567e+04	-5.242800e+05	-7.411664e+05	1.346264e+06
63	-8.245126e+05	-9.692727e+05	9.692727e+05	-2.506341e+04	-5.728917e+05	-7.814452e+05	1.334089e+06
64	-8.366863e+05	-1.030231e+06	1.030231e+06	-3.092584e+04	-6.228445e+05	-8.200514e+05	1.318636e+06
65	-8.485056e+05	-1.090818e+06	1.090818e+06	-3.727282e+04	-6.740144e+05	-8.568540e+05	1.299986e+06
66	-8.600104e+05	-1.150919e+06	1.150919e+06	-4.409222e+04	-7.262762e+05	-8.917356e+05	1.278225e+06
67	-8.712391e+05	-1.210427e+06	1.210427e+06	-5.137009e+04	-7.795035e+05	-9.245913e+05	1.253434e+06
68	-8.822290e+05	-1.269245e+06	1.269245e+06	-5.909083e+04	-8.335693e+05	-9.553287e+05	1.225696e+06
69	-8.930157e+05	-1.327280e+06	1.327280e+06	-6.723730e+04	-8.883460e+05	-9.838662e+05	1.195088e+06
70	-9.036332e+05	-1.384446e+06	1.384446e+06	-7.579097e+04	-9.437052e+05	-1.010133e+06	1.161688e+06
71	-9.141136e+05	-1.440663e+06	1.440663e+06	-8.473195e+04	-9.995178e+05	-1.034067e+06	1.125569e+06
72	-9.244869e+05	-1.495851e+06	1.495851e+06	-9.403911e+04	-1.055654e+06	-1.055615e+06	1.086800e+06
73	-9.347811e+05	-1.549937e+06	1.549937e+06	-1.036901e+05	-1.111983e+06	-1.074732e+06	1.045446e+06
74	-9.450217e+05	-1.602848e+06	1.602848e+06	-1.136612e+05	-1.168371e+06	-1.091380e+06	1.001570e+06
75	-9.552318e+05	-1.654510e+06	1.654510e+06	-1.239277e+05	-1.224685e+06	-1.105528e+06	9.552318e+05
76	-9.654321e+05	-1.704852e+06	1.704852e+06	-1.344635e+05	-1.280787e+06	-1.117150e+06	9.064858e+05
77	-9.756403e+05	-1.753799e+06	1.753799e+06	-1.452411e+05	-1.336537e+06	-1.126226e+06	8.553851e+05
78	-9.858717e+05	-1.801277e+06	1.801277e+06	-1.562320e+05	-1.391793e+06	-1.132742e+06	8.019803e+05
79	-9.961384e+05	-1.847207e+06	1.847207e+06	-1.674057e+05	-1.446407e+06	-1.136687e+06	7.463200e+05
80	-1.006450e+06	-1.891507e+06	1.891507e+06	-1.787308e+05	-1.500227e+06	-1.138056e+06	6.884522e+05
81	-1.016812e+06	-1.934091e+06	1.934091e+06	-1.901740e+05	-1.553099e+06	-1.136850e+06	6.284244e+05
82	-1.027228e+06	-1.974871e+06	1.974871e+06	-2.017002e+05	-1.604861e+06	-1.133072e+06	5.662851e+05
83	-1.037699e+06	-2.013750e+06	2.013750e+06	-2.132730e+05	-1.655348e+06	-1.126734e+06	5.020843e+05
84	-1.048221e+06	-2.050630e+06	2.050630e+06	-2.248538e+05	-1.704388e+06	-1.117849e+06	4.358749e+05
85	-1.058788e+06	-2.085406e+06	2.085406e+06	-2.364024e+05	-1.751805e+06	-1.106439e+06	3.677133e+05
86	-1.069391e+06	-2.117967e+06	2.117967e+06	-2.478767e+05	-1.797419e+06	-1.092532e+06	2.976609e+05
87	-1.080017e+06	-2.148200e+06	2.148200e+06	-2.592327e+05	-1.841043e+06	-1.076162e+06	2.257849e+05
88	-1.090650e+06	-2.175986e+06	2.175986e+06	-2.704244e+05	-1.882486e+06	-1.057371e+06	1.521598e+05
89	-1.101272e+06	-2.201202e+06	2.201202e+06	-2.814044e+05	-1.921555e+06	-1.036207e+06	7.686766e+04
90	-1.111861e+06	-2.223721e+06	2.223721e+06	-2.921232e+05	-1.958054e+06	-1.012732e+06	2.719781e-10
91	-1.122391e+06	-2.243415e+06	2.243415e+06	-3.025299e+05	-1.991782e+06	-9.870108e+05	-7.834177e+04
92	-1.132836e+06	-2.260153e+06	2.260153e+06	-3.125721e+05	-2.022541e+06	-9.591232e+05	-1.580453e+05
93	-1.143164e+06	-2.273803e+06	2.273803e+06	-3.221962e+05	-2.050131e+06	-9.291576e+05	-2.389863e+05
94	-1.153342e+06	-2.284235e+06	2.284235e+06	-3.313476e+05	-2.074354e+06	-8.972138e+05	-3.210283e+05
95	-1.163333e+06	-2.291319e+06	2.291319e+06	-3.399710e+05	-2.095017e+06	-8.634032e+05	-4.040214e+05
96	-1.173100e+06	-2.294929e+06	2.294929e+06	-3.480106e+05	-2.111931e+06	-8.278492e+05	-4.878022e+05
97	-1.182599e+06	-2.294942e+06	2.294942e+06	-3.554106e+05	-2.124913e+06	-7.906874e+05	-5.721934e+05
98	-1.191789e+06	-2.291243e+06	2.291243e+06	-3.621157e+05	-2.133791e+06	-7.520656e+05	-6.570033e+05
99	-1.200623e+06	-2.283721e+06	2.283721e+06	-3.680712e+05	-2.138401e+06	-7.121440e+05	-7.420261e+05

deg	B (m/s)	F_lat signed (N)	|F_lat| (N)	F_lat x	F_lat y	F_lat z	F_parallel (N)
100	-1.209054e+06	-2.272278e+06	2.272278e+06	-3.732237e+05	-2.138594e+06	-6.710945e+05	-8.270417e+05
101	-1.217032e+06	-2.256824e+06	2.256824e+06	-3.775217e+05	-2.134236e+06	-6.291011e+05	-9.118161e+05
102	-1.224504e+06	-2.237281e+06	2.237281e+06	-3.809157e+05	-2.125209e+06	-5.863583e+05	-9.961017e+05
103	-1.231420e+06	-2.213586e+06	2.213586e+06	-3.833591e+05	-2.111415e+06	-5.430715e+05	-1.079638e+06
104	-1.237724e+06	-2.185691e+06	2.185691e+06	-3.848089e+05	-2.092776e+06	-4.994549e+05	-1.162152e+06
105	-1.243361e+06	-2.153564e+06	2.153564e+06	-3.852258e+05	-2.069239e+06	-4.557314e+05	-1.243361e+06
106	-1.248274e+06	-2.117194e+06	2.117194e+06	-3.845750e+05	-2.040774e+06	-4.121305e+05	-1.322969e+06
107	-1.252408e+06	-2.076586e+06	2.076586e+06	-3.828268e+05	-2.007380e+06	-3.688871e+05	-1.400675e+06
108	-1.255703e+06	-2.031770e+06	2.031770e+06	-3.799572e+05	-1.969084e+06	-3.262399e+05	-1.476167e+06
109	-1.258102e+06	-1.982797e+06	1.982797e+06	-3.759481e+05	-1.925940e+06	-2.844293e+05	-1.549130e+06
110	-1.259548e+06	-1.929740e+06	1.929740e+06	-3.707884e+05	-1.878037e+06	-2.436954e+05	-1.619244e+06
111	-1.259982e+06	-1.872698e+06	1.872698e+06	-3.644739e+05	-1.825494e+06	-2.042761e+05	-1.686185e+06
112	-1.259346e+06	-1.811796e+06	1.811796e+06	-3.570081e+05	-1.768462e+06	-1.664045e+05	-1.749631e+06
113	-1.257584e+06	-1.747183e+06	1.747183e+06	-3.484022e+05	-1.707128e+06	-1.303065e+05	-1.809261e+06
114	-1.254640e+06	-1.679036e+06	1.679036e+06	-3.386761e+05	-1.641708e+06	-9.619884e+04	-1.864758e+06
115	-1.250458e+06	-1.607558e+06	1.607558e+06	-3.278581e+05	-1.572456e+06	-6.428591e+04	-1.915813e+06
116	-1.244984e+06	-1.532977e+06	1.532977e+06	-3.159851e+05	-1.499655e+06	-3.475771e+04	-1.962122e+06
117	-1.238166e+06	-1.455551e+06	1.455551e+06	-3.031032e+05	-1.423621e+06	-7.787146e+03	-2.003395e+06
118	-1.229953e+06	-1.375562e+06	1.375562e+06	-2.892675e+05	-1.344702e+06	1.647241e+04	-2.039355e+06
119	-1.220297e+06	-1.293317e+06	1.293317e+06	-2.745418e+05	-1.263274e+06	3.788954e+04	-2.069740e+06
120	-1.209150e+06	-1.209150e+06	1.209150e+06	-2.589992e+05	-1.179740e+06	5.635704e+04	-2.094308e+06
121	-1.196468e+06	-1.123415e+06	1.123415e+06	-2.427210e+05	-1.094529e+06	7.179416e+04	-2.112836e+06
122	-1.182209e+06	-1.036493e+06	1.036493e+06	-2.257974e+05	-1.008093e+06	8.414853e+04	-2.125125e+06
123	-1.166335e+06	-9.487824e+05	9.487824e+05	-2.083264e+05	-9.209045e+05	9.339802e+04	-2.131000e+06
124	-1.148809e+06	-8.607030e+05	8.607030e+05	-1.904136e+05	-8.334517e+05	9.955229e+04	-2.130315e+06
125	-1.129599e+06	-7.726911e+05	7.726911e+05	-1.721716e+05	-7.462376e+05	1.026541e+05	-2.122951e+06
126	-1.108674e+06	-6.851984e+05	6.851984e+05	-1.537197e+05	-6.597751e+05	1.027805e+05	-2.108824e+06
127	-1.086009e+06	-5.986895e+05	5.986895e+05	-1.351827e+05	-5.745834e+05	1.000433e+05	-2.087879e+06
128	-1.061582e+06	-5.136399e+05	5.136399e+05	-1.166904e+05	-4.911844e+05	9.458962e+04	-2.060097e+06
129	-1.035374e+06	-4.305326e+05	4.305326e+05	-9.837652e+04	-4.100981e+05	8.660214e+04	-2.025497e+06
130	-1.007370e+06	-3.498559e+05	3.498559e+05	-8.037809e+04	-3.318389e+05	7.629852e+04	-1.984131e+06
131	-9.775601e+05	-2.721001e+05	2.721001e+05	-6.283419e+04	-2.569109e+05	6.393089e+04	-1.936093e+06
132	-9.459385e+05	-1.977550e+05	1.977550e+05	-4.588498e+04	-1.858039e+05	4.978480e+04	-1.881513e+06
133	-9.125032e+05	-1.273060e+05	1.273060e+05	-2.967059e+04	-1.189889e+05	3.417780e+04	-1.820561e+06
134	-8.772572e+05	-6.123167e+04	6.123167e+04	-1.433004e+04	-5.691396e+04	1.745761e+04	-1.753446e+06
135	-8.402079e+05	-3.086874e-10	3.406516e-10	-2.328306e-10	-2.328306e-10	8.731149e-11	-1.680416e+06
136	-8.013674e+05	5.593464e+04	5.593464e+04	1.318632e+04	5.136327e+04	-1.779381e+04	-1.601759e+06
137	-7.607526e+05	1.061348e+05	1.061348e+05	2.510038e+04	9.682110e+04	-3.549999e+04	-1.517799e+06
138	-7.183851e+05	1.501834e+05	1.501834e+05	3.561948e+04	1.360574e+05	-5.267540e+04	-1.428899e+06
139	-6.742916e+05	1.876865e+05	1.876865e+05	4.462792e+04	1.687979e+05	-6.886106e+04	-1.335459e+06
140	-6.285033e+05	2.182769e+05	2.182769e+05	5.201809e+04	1.948135e+05	-8.358606e+04	-1.237910e+06
141	-5.810567e+05	2.416170e+05	2.416170e+05	5.769165e+04	2.139228e+05	-9.637160e+04	-1.136718e+06
142	-5.319930e+05	2.574015e+05	2.574015e+05	6.156045e+04	2.259943e+05	-1.067353e+05	-1.032381e+06
143	-4.813583e+05	2.653607e+05	2.653607e+05	6.354761e+04	2.309487e+05	-1.141959e+05	-9.254226e+05
144	-4.292037e+05	2.652625e+05	2.652625e+05	6.358834e+04	2.287600e+05	-1.182774e+05	-8.163939e+05
145	-3.755849e+05	2.569152e+05	2.569152e+05	6.163083e+04	2.194569e+05	-1.185147e+05	-7.058688e+05
146	-3.205627e+05	2.401698e+05	2.401698e+05	5.763690e+04	2.031232e+05	-1.144574e+05	-5.944412e+05
147	-2.642024e+05	2.149216e+05	2.149216e+05	5.158267e+04	1.798977e+05	-1.056755e+05	-4.827217e+05
148	-2.065738e+05	1.811120e+05	1.811120e+05	4.345908e+04	1.499743e+05	-9.176375e+04	-3.713347e+05
149	-1.477517e+05	1.387304e+05	1.387304e+05	3.327224e+04	1.136006e+05	-7.234641e+04	-2.609140e+05
150	-8.781484e+04	8.781484e+04	8.781484e+04	2.104373e+04	7.107667e+04	-4.708201e+04	-1.520998e+05
151	-2.684655e+04	2.845301e+04	2.845301e+04	6.810748e+03	2.275330e+04	-1.566763e+04	-4.553433e+04

deg	B (m/s)	F_lat signed (N)	|F_lat| (N)	F_lat x	F_lat y	F_lat z	F_parallel (N)
152	3.506576e+04	-3.921705e+04	3.921705e+04	-9.373901e+03	-3.097064e+04	2.215685e+04	5.814167e+04
153	9.783065e+04	-1.150068e+05	1.150068e+05	-2.744192e+04	-8.965097e+04	6.660491e+04	1.582933e+05
154	1.613528e+05	-1.986774e+05	1.986774e+05	-4.730978e+04	-1.528015e+05	1.178398e+05	2.542955e+05
155	2.255333e+05	-2.899400e+05	2.899400e+05	-6.887913e+04	-2.198977e+05	1.759711e+05	3.455370e+05
156	2.902695e+05	-3.884564e+05	3.884564e+05	-9.203733e+04	-2.903812e+05	2.410525e+05	4.314245e+05
157	3.554558e+05	-4.938406e+05	4.938406e+05	-1.166581e+05	-3.636636e+05	3.130786e+05	5.113870e+05
158	4.209832e+05	-6.056599e+05	6.056599e+05	-1.426024e+05	-4.391321e+05	3.919840e+05	5.848790e+05
159	4.867400e+05	-7.234366e+05	7.234366e+05	-1.697191e+05	-5.161541e+05	4.776410e+05	6.513853e+05
160	5.526118e+05	-8.466505e+05	8.466505e+05	-1.978464e+05	-5.940825e+05	5.698595e+05	7.104241e+05
161	6.184820e+05	-9.747409e+05	9.747409e+05	-2.268128e+05	-6.722616e+05	6.683862e+05	7.615510e+05
162	6.842314e+05	-1.107110e+06	1.107110e+06	-2.564381e+05	-7.500326e+05	7.729053e+05	8.043623e+05
163	7.497395e+05	-1.243124e+06	1.243124e+06	-2.865352e+05	-8.267390e+05	8.830393e+05	8.384980e+05
164	8.148836e+05	-1.382121e+06	1.382121e+06	-3.169111e+05	-9.017331e+05	9.983502e+05	8.636450e+05
165	8.795401e+05	-1.523408e+06	1.523408e+06	-3.473688e+05	-9.743810e+05	1.118342e+06	8.795401e+05
166	9.435841e+05	-1.666271e+06	1.666271e+06	-3.777083e+05	-1.044069e+06	1.242463e+06	8.859718e+05
167	1.006890e+06	-1.809974e+06	1.809974e+06	-4.077288e+05	-1.110208e+06	1.370110e+06	8.827831e+05
168	1.069332e+06	-1.953766e+06	1.953766e+06	-4.372299e+05	-1.172240e+06	1.500629e+06	8.698729e+05
169	1.130783e+06	-2.096888e+06	2.096888e+06	-4.660132e+05	-1.229643e+06	1.633324e+06	8.471976e+05
170	1.191118e+06	-2.238569e+06	2.238569e+06	-4.938841e+05	-1.281936e+06	1.767459e+06	8.147725e+05
171	1.250210e+06	-2.378041e+06	2.378041e+06	-5.206535e+05	-1.328681e+06	1.902263e+06	7.726723e+05
172	1.307935e+06	-2.514535e+06	2.514535e+06	-5.461389e+05	-1.369490e+06	2.036938e+06	7.210313e+05
173	1.364168e+06	-2.647293e+06	2.647293e+06	-5.701666e+05	-1.404029e+06	2.170662e+06	6.600442e+05
174	1.418787e+06	-2.775566e+06	2.775566e+06	-5.925726e+05	-1.432017e+06	2.302597e+06	5.899648e+05
175	1.471671e+06	-2.898626e+06	2.898626e+06	-6.132040e+05	-1.453234e+06	2.431897e+06	5.111059e+05
176	1.522700e+06	-3.015763e+06	3.015763e+06	-6.319208e+05	-1.467518e+06	2.557712e+06	4.238379e+05
177	1.571759e+06	-3.126297e+06	3.126297e+06	-6.485965e+05	-1.474770e+06	2.679199e+06	3.285870e+05
178	1.618731e+06	-3.229576e+06	3.229576e+06	-6.631197e+05	-1.474954e+06	2.795523e+06	2.258340e+05
179	1.663507e+06	-3.324987e+06	3.324987e+06	-6.753945e+05	-1.468095e+06	2.905870e+06	1.161111e+05
180	1.705977e+06	-3.411954e+06	3.411954e+06	-6.853418e+05	-1.454282e+06	3.009452e+06	1.087542e-09
181	1.746037e+06	-3.489947e+06	3.489947e+06	-6.928998e+05	-1.433668e+06	3.105514e+06	-1.218716e+05
182	1.783586e+06	-3.558482e+06	3.558482e+06	-6.980244e+05	-1.406461e+06	3.193341e+06	-2.488333e+05
183	1.818526e+06	-3.617128e+06	3.617128e+06	-7.006897e+05	-1.372931e+06	3.272263e+06	-3.801754e+05
184	1.850764e+06	-3.665506e+06	3.665506e+06	-7.008883e+05	-1.333403e+06	3.341665e+06	-5.151532e+05
185	1.880213e+06	-3.703296e+06	3.703296e+06	-6.986311e+05	-1.288252e+06	3.400989e+06	-6.529910e+05
186	1.906787e+06	-3.730239e+06	3.730239e+06	-6.939473e+05	-1.237902e+06	3.449741e+06	-7.928867e+05
187	1.930410e+06	-3.746136e+06	3.746136e+06	-6.868844e+05	-1.182823e+06	3.487500e+06	-9.340167e+05
188	1.951006e+06	-3.750855e+06	3.750855e+06	-6.775070e+05	-1.123521e+06	3.513915e+06	-1.075540e+06
189	1.968509e+06	-3.744327e+06	3.744327e+06	-6.658970e+05	-1.060538e+06	3.528714e+06	-1.216605e+06
190	1.982856e+06	-3.726550e+06	3.726550e+06	-6.521525e+05	-9.944432e+05	3.531707e+06	-1.356353e+06
191	1.993990e+06	-3.697591e+06	3.697591e+06	-6.363867e+05	-9.258299e+05	3.522787e+06	-1.493924e+06
192	2.001862e+06	-3.657584e+06	3.657584e+06	-6.187272e+05	-8.553068e+05	3.501935e+06	-1.628461e+06
193	2.006427e+06	-3.606729e+06	3.606729e+06	-5.993144e+05	-7.834933e+05	3.469215e+06	-1.759119e+06
194	2.007647e+06	-3.545295e+06	3.545295e+06	-5.783007e+05	-7.110125e+05	3.424784e+06	-1.885067e+06
195	2.005492e+06	-3.473614e+06	3.473614e+06	-5.558488e+05	-6.384849e+05	3.368881e+06	-2.005492e+06
196	1.999936e+06	-3.392084e+06	3.392084e+06	-5.321301e+05	-5.665220e+05	3.301836e+06	-2.119609e+06
197	1.990962e+06	-3.301165e+06	3.301165e+06	-5.073233e+05	-4.957196e+05	3.224062e+06	-2.226664e+06
198	1.978559e+06	-3.201376e+06	3.201376e+06	-4.816127e+05	-4.266519e+05	3.136052e+06	-2.325936e+06
199	1.962723e+06	-3.093294e+06	3.093294e+06	-4.551863e+05	-3.598651e+05	3.038383e+06	-2.416746e+06
200	1.943457e+06	-2.977550e+06	2.977550e+06	-4.282346e+05	-2.958716e+05	2.931702e+06	-2.498461e+06
201	1.920772e+06	-2.854823e+06	2.854823e+06	-4.009481e+05	-2.351448e+05	2.816729e+06	-2.570495e+06
202	1.894684e+06	-2.725843e+06	2.725843e+06	-3.735163e+05	-1.781134e+05	2.694250e+06	-2.632316e+06
203	1.865218e+06	-2.591378e+06	2.591378e+06	-3.461252e+05	-1.251573e+05	2.565107e+06	-2.683451e+06

deg	B (m/s)	F_lat signed (N)	|F_lat| (N)	F_lat x	F_lat y	F_lat z	F_parallel (N)
204	1.832405e+06	-2.452236e+06	2.452236e+06	-3.189562e+05	-7.660274e+04	2.430198e+06	-2.723484e+06
205	1.796284e+06	-2.309258e+06	2.309258e+06	-2.921844e+05	-3.271928e+04	2.290465e+06	-2.752066e+06
206	1.756900e+06	-2.163312e+06	2.163312e+06	-2.659764e+05	6.283860e+03	2.146889e+06	-2.768913e+06
207	1.714307e+06	-2.015289e+06	2.015289e+06	-2.404897e+05	4.026011e+04	2.000483e+06	-2.773807e+06
208	1.668564e+06	-1.866098e+06	1.866098e+06	-2.158706e+05	6.912747e+04	1.852281e+06	-2.766604e+06
209	1.619737e+06	-1.716659e+06	1.716659e+06	-1.922530e+05	9.286971e+04	1.703330e+06	-2.747229e+06
210	1.567899e+06	-1.567899e+06	1.567899e+06	-1.697575e+05	1.115370e+05	1.554687e+06	-2.715681e+06
211	1.513132e+06	-1.420744e+06	1.420744e+06	-1.484904e+05	1.252456e+05	1.407402e+06	-2.672032e+06
212	1.455520e+06	-1.276116e+06	1.276116e+06	-1.285422e+05	1.341774e+05	1.262515e+06	-2.616425e+06
213	1.395156e+06	-1.134922e+06	1.134922e+06	-1.099876e+05	1.385778e+05	1.121048e+06	-2.549078e+06
214	1.332141e+06	-9.980575e+05	9.980575e+05	-9.288436e+04	1.387543e+05	9.839911e+05	-2.470279e+06
215	1.266578e+06	-8.663904e+05	8.663904e+05	-7.727332e+04	1.350729e+05	8.523007e+05	-2.380388e+06
216	1.198579e+06	-7.407623e+05	7.407623e+05	-6.317785e+04	1.279550e+05	7.268871e+05	-2.279832e+06
217	1.128259e+06	-6.219807e+05	6.219807e+05	-5.060392e+04	1.178733e+05	6.086092e+05	-2.169105e+06
218	1.055741e+06	-5.108139e+05	5.108139e+05	-3.954016e+04	1.053470e+05	4.982664e+05	-2.048763e+06
219	9.811523e+05	-4.079861e+05	4.079861e+05	-2.995822e+04	9.093665e+04	3.965926e+05	-1.919423e+06
220	9.046237e+05	-3.141725e+05	3.141725e+05	-2.181314e+04	7.523856e+04	3.042494e+05	-1.781761e+06
221	8.262921e+05	-2.299953e+05	2.299953e+05	-1.504405e+04	5.887852e+04	2.218216e+05	-1.636501e+06
222	7.462983e+05	-1.560188e+05	1.560188e+05	-9.574896e+03	4.250546e+04	1.498115e+05	-1.484420e+06
223	6.647872e+05	-9.274643e+04	9.274643e+04	-5.315420e+03	2.678463e+04	8.863537e+04	-1.326336e+06
224	5.819075e+05	-4.061656e+04	4.061656e+04	-2.162224e+03	1.239059e+04	3.861998e+04	-1.163106e+06
225	4.978113e+05	-3.048215e-10	3.214623e-10	-1.164153e-10	1.455192e-10	2.619345e-10	-9.956225e+05
226	4.126538e+05	2.880282e+04	2.880282e+04	1.297137e+03	-9.715628e+03	-2.708369e+04	-8.248048e+05
227	3.265931e+05	4.556396e+04	4.556396e+04	1.864200e+03	-1.609729e+04	-4.258494e+04	-6.515950e+05
228	2.397898e+05	5.012971e+04	5.012971e+04	1.843784e+03	-1.850573e+04	-4.655240e+04	-4.769524e+05
229	1.524065e+05	4.242178e+04	4.242178e+04	1.384453e+03	-1.632864e+04	-3.912885e+04	-3.018467e+05
230	6.460790e+04	2.243809e+04	2.243809e+04	6.390509e+02	-8.987554e+03	-2.054953e+04	-1.272527e+05
231	-2.344021e+04	-9.746989e+03	9.746989e+03	-2.370192e+02	4.055375e+03	8.860110e+03	4.585598e+04
232	-1.115708e+05	-5.398282e+04	5.398282e+04	-1.087554e+03	2.329105e+04	4.868767e+04	2.165133e+05
233	-1.996160e+05	-1.100433e+05	1.100433e+05	-1.757313e+03	4.915745e+04	9.843767e+04	3.837664e+05
234	-2.874078e+05	-1.776278e+05	1.776278e+05	-2.093773e+03	8.203408e+04	1.575362e+05	5.466821e+05
235	-3.747780e+05	-2.563632e+05	2.563632e+05	-1.948863e+03	1.222370e+05	2.253363e+05	7.043522e+05
236	-4.615588e+05	-3.458060e+05	3.458060e+05	-1.180644e+03	1.700145e+05	3.011237e+05	8.558998e+05
237	-5.475833e+05	-4.454444e+05	4.454444e+05	3.450575e+02	2.255430e+05	3.841236e+05	1.000484e+06
238	-6.326854e+05	-5.547021e+05	5.547021e+05	2.753107e+03	2.889242e+05	4.735078e+05	1.137308e+06
239	-7.167007e+05	-6.729412e+05	6.729412e+05	6.157605e+03	3.601826e+05	5.684017e+05	1.265618e+06
240	-7.994665e+05	-7.994665e+05	7.994665e+05	1.066054e+04	4.392635e+05	6.678926e+05	1.384717e+06
241	-8.808222e+05	-9.335293e+05	9.335293e+05	1.635057e+04	5.260326e+05	7.710379e+05	1.493959e+06
242	-9.606100e+05	-1.074333e+06	1.074333e+06	2.330192e+04	6.202751e+05	8.768731e+05	1.592764e+06
243	-1.038675e+06	-1.221035e+06	1.221035e+06	3.157347e+04	7.216967e+05	9.844206e+05	1.680611e+06
244	-1.114864e+06	-1.372758e+06	1.372758e+06	4.120793e+04	8.299252e+05	1.092699e+06	1.757050e+06
245	-1.189030e+06	-1.528588e+06	1.528588e+06	5.223126e+04	9.445119e+05	1.200729e+06	1.821700e+06
246	-1.261029e+06	-1.687586e+06	1.687586e+06	6.465220e+04	1.064935e+06	1.307547e+06	1.874254e+06
247	-1.330719e+06	-1.848790e+06	1.848790e+06	7.846200e+04	1.190603e+06	1.412209e+06	1.914479e+06
248	-1.397966e+06	-2.011225e+06	2.011225e+06	9.363436e+04	1.320860e+06	1.513798e+06	1.942217e+06
249	-1.462637e+06	-2.173902e+06	2.173902e+06	1.101255e+05	1.454989e+06	1.611438e+06	1.957390e+06
250	-1.524608e+06	-2.335835e+06	2.335835e+06	1.278743e+05	1.592217e+06	1.704293e+06	1.959998e+06
251	-1.583757e+06	-2.496035e+06	2.496035e+06	1.468032e+05	1.731725e+06	1.791583e+06	1.950117e+06
252	-1.639970e+06	-2.653528e+06	2.653528e+06	1.668183e+05	1.872651e+06	1.872581e+06	1.927901e+06
253	-1.693138e+06	-2.807350e+06	2.807350e+06	1.878104e+05	2.014098e+06	1.946626e+06	1.893582e+06
254	-1.743158e+06	-2.956564e+06	2.956564e+06	2.096559e+05	2.155142e+06	2.013126e+06	1.847466e+06
255	-1.789933e+06	-3.100255e+06	3.100255e+06	2.322183e+05	2.294840e+06	2.071561e+06	1.789933e+06

deg	B (m/s)	F_lat signed (N)	|F_lat| (N)	F_lat x	F_lat y	F_lat z	F_parallel (N)
256	-1.833374e+06	-3.237546e+06	3.237546e+06	2.553487e+05	2.432239e+06	2.121489e+06	1.721434e+06
257	-1.873396e+06	-3.367595e+06	3.367595e+06	2.788878e+05	2.566380e+06	2.162547e+06	1.642486e+06
258	-1.909924e+06	-3.489606e+06	3.489606e+06	3.026674e+05	2.696314e+06	2.194455e+06	1.553673e+06
259	-1.942889e+06	-3.602831e+06	3.602831e+06	3.265117e+05	2.821102e+06	2.217017e+06	1.455638e+06
260	-1.972229e+06	-3.706578e+06	3.706578e+06	3.502392e+05	2.939830e+06	2.230123e+06	1.349084e+06
261	-1.997888e+06	-3.800209e+06	3.800209e+06	3.736643e+05	3.051614e+06	2.233745e+06	1.234763e+06
262	-2.019821e+06	-3.883152e+06	3.883152e+06	3.965995e+05	3.155609e+06	2.227939e+06	1.113476e+06
263	-2.037986e+06	-3.954899e+06	3.954899e+06	4.188569e+05	3.251015e+06	2.212845e+06	9.860671e+05
264	-2.052354e+06	-4.015010e+06	4.015010e+06	4.402502e+05	3.337088e+06	2.188681e+06	8.534166e+05
265	-2.062898e+06	-4.063116e+06	4.063116e+06	4.605966e+05	3.413144e+06	2.155740e+06	7.164371e+05
266	-2.069604e+06	-4.098925e+06	4.098925e+06	4.797186e+05	3.478565e+06	2.114390e+06	5.760664e+05
267	-2.072462e+06	-4.122218e+06	4.122218e+06	4.974459e+05	3.532808e+06	2.065066e+06	4.332626e+05
268	-2.071472e+06	-4.132852e+06	4.132852e+06	5.136173e+05	3.575407e+06	2.008265e+06	2.889972e+05
269	-2.066641e+06	-4.130764e+06	4.130764e+06	5.280820e+05	3.605981e+06	1.944541e+06	1.442494e+05
270	-2.057983e+06	-4.115966e+06	4.115966e+06	5.407014e+05	3.624233e+06	1.874502e+06	1.507403e-09
271	-2.045521e+06	-4.088551e+06	4.088551e+06	5.513508e+05	3.629958e+06	1.798795e+06	-1.427753e+05
272	-2.029286e+06	-4.048686e+06	4.048686e+06	5.599206e+05	3.623044e+06	1.718109e+06	-2.831117e+05
273	-2.009315e+06	-3.996615e+06	3.996615e+06	5.663174e+05	3.603471e+06	1.633161e+06	-4.200612e+05
274	-1.985653e+06	-3.932658e+06	3.932658e+06	5.704652e+05	3.571316e+06	1.544690e+06	-5.526990e+05
275	-1.958353e+06	-3.857203e+06	3.857203e+06	5.723066e+05	3.526749e+06	1.453451e+06	-6.801289e+05
276	-1.927475e+06	-3.770711e+06	3.770711e+06	5.718029e+05	3.470034e+06	1.360208e+06	-8.014893e+05
277	-1.893086e+06	-3.673707e+06	3.673707e+06	5.689356e+05	3.401527e+06	1.265720e+06	-9.159580e+05
278	-1.855260e+06	-3.566780e+06	3.566780e+06	5.637060e+05	3.321675e+06	1.170741e+06	-1.022758e+06
279	-1.814077e+06	-3.450579e+06	3.450579e+06	5.561356e+05	3.231008e+06	1.076011e+06	-1.121161e+06
280	-1.769624e+06	-3.325806e+06	3.325806e+06	5.462666e+05	3.130140e+06	9.822434e+05	-1.210494e+06
281	-1.721996e+06	-3.193214e+06	3.193214e+06	5.341611e+05	3.019763e+06	8.901246e+05	-1.290142e+06
282	-1.671292e+06	-3.053602e+06	3.053602e+06	5.199011e+05	2.900638e+06	8.003040e+05	-1.359551e+06
283	-1.617616e+06	-2.907808e+06	2.907808e+06	5.035878e+05	2.773594e+06	7.133889e+05	-1.418233e+06
284	-1.561082e+06	-2.756707e+06	2.756707e+06	4.853411e+05	2.639518e+06	6.299385e+05	-1.465767e+06
285	-1.501804e+06	-2.601201e+06	2.601201e+06	4.652984e+05	2.499348e+06	5.504591e+05	-1.501804e+06
286	-1.439905e+06	-2.442218e+06	2.442218e+06	4.436137e+05	2.354067e+06	4.753993e+05	-1.526067e+06
287	-1.375512e+06	-2.280702e+06	2.280702e+06	4.204563e+05	2.204693e+06	4.051464e+05	-1.538353e+06
288	-1.308754e+06	-2.117609e+06	2.117609e+06	3.960097e+05	2.052274e+06	3.400229e+05	-1.538533e+06
289	-1.239768e+06	-1.953901e+06	1.953901e+06	3.704695e+05	1.897874e+06	2.802843e+05	-1.526555e+06
290	-1.168693e+06	-1.790541e+06	1.790541e+06	3.440423e+05	1.742568e+06	2.261169e+05	-1.502443e+06
291	-1.095671e+06	-1.628484e+06	1.628484e+06	3.169438e+05	1.587436e+06	1.776370e+05	-1.466294e+06
292	-1.020848e+06	-1.468674e+06	1.468674e+06	2.893970e+05	1.433547e+06	1.348904e+05	-1.418282e+06
293	-9.443739e+05	-1.312034e+06	1.312034e+06	2.616301e+05	1.281955e+06	9.785275e+04	-1.358651e+06
294	-8.663993e+05	-1.159469e+06	1.159469e+06	2.338749e+05	1.133692e+06	6.643070e+04	-1.287720e+06
295	-7.870783e+05	-1.011848e+06	1.011848e+06	2.063644e+05	9.897543e+05	4.046362e+04	-1.205874e+06
296	-7.065665e+05	-8.700116e+05	8.700116e+05	1.793312e+05	8.511001e+05	1.972607e+04	-1.113564e+06
297	-6.250212e+05	-7.347565e+05	7.347565e+05	1.530053e+05	7.186384e+05	3.930920e+03	-1.011306e+06
298	-5.426008e+05	-6.068370e+05	6.068370e+05	1.276120e+05	5.932230e+05	-7.266899e+03	-8.996729e+05
299	-4.594646e+05	-4.869582e+05	4.869582e+05	1.033702e+05	4.756463e+05	-1.426612e+04	-7.792961e+05
300	-3.757722e+05	-3.757722e+05	3.757722e+05	8.049020e+04	3.666324e+05	-1.751430e+04	-6.508566e+05
301	-2.916837e+05	-2.738744e+05	2.738744e+05	5.917232e+04	2.668323e+05	-1.750251e+04	-5.150829e+05
302	-2.073587e+05	-1.818002e+05	1.818002e+05	3.960472e+04	1.768189e+05	-1.475960e+04	-3.727456e+05
303	-1.229564e+05	-1.000218e+05	1.000218e+05	2.196201e+04	9.708284e+04	-9.846130e+03	-2.246526e+05
304	-3.863501e+04	-2.894586e+04	2.894586e+04	6.403700e+03	2.802939e+04	-3.347992e+03	-7.164352e+04
305	4.544851e+04	3.108861e+04	3.108861e+04	-6.927188e+03	-3.002427e+04	4.130207e+03	8.541525e+04
306	1.291387e+05	7.981212e+04	7.981212e+04	-1.790532e+04	-7.685082e+04	1.197191e+04	2.456365e+05
307	2.122822e+05	1.170258e+05	1.170258e+05	-2.642416e+04	-1.123138e+05	1.955545e+04	4.081174e+05

deg	B (m/s)	F_lat signed (N)	|F_lat| (N)	F_lat x	F_lat y	F_lat z	F_parallel (N)
308	2.947276e+05	1.426021e+05	1.426021e+05	-3.239681e+04	-1.363678e+05	2.626096e+04	5.719458e+05
309	3.763262e+05	1.564852e+05	1.564852e+05	-3.575681e+04	-1.490579e+05	3.147719e+04	7.362052e+05
310	4.569323e+05	1.586909e+05	1.586909e+05	-3.645865e+04	-1.505186e+05	3.460820e+04	8.999810e+05
311	5.364032e+05	1.493058e+05	1.493058e+05	-3.447814e+04	-1.409712e+05	3.507992e+04	1.062366e+06
312	6.145996e+05	1.284863e+05	1.284863e+05	-2.981260e+04	-1.207214e+05	3.234642e+04	1.222466e+06
313	6.913861e+05	9.645732e+04	9.645732e+04	-2.248084e+04	-9.015559e+04	2.589586e+04	1.379404e+06
314	7.666313e+05	5.351009e+04	5.351009e+04	-1.252296e+04	-4.973687e+04	1.525613e+04	1.532329e+06
315	8.402079e+05	7.202706e-10	7.459897e-10	0.000000e+00	-6.984919e-10	2.619345e-10	1.680416e+06
316	9.119934e+05	-6.365622e+04	6.365622e+04	1.500665e+04	5.845379e+04	-2.025018e+04	1.822876e+06
317	9.818697e+05	-1.369835e+05	1.369835e+05	3.239595e+04	1.249627e+05	-4.581827e+04	1.958956e+06
318	1.049724e+06	-2.194521e+05	2.194521e+05	5.204817e+04	1.988107e+05	-7.697073e+04	2.087947e+06
319	1.115449e+06	-3.104809e+05	3.104809e+05	7.382584e+04	2.792343e+05	-1.139136e+05	2.209186e+06
320	1.178941e+06	-4.094419e+05	4.094419e+05	9.757507e+04	3.654294e+05	-1.567900e+05	2.322060e+06
321	1.240104e+06	-5.156643e+05	5.156643e+05	1.231268e+05	4.565587e+05	-2.056784e+05	2.426010e+06
322	1.298848e+06	-6.284393e+05	6.284393e+05	1.502983e+05	5.517594e+05	-2.605916e+05	2.520532e+06
323	1.355086e+06	-7.470244e+05	7.470244e+05	1.788947e+05	6.501502e+05	-3.214760e+05	2.605184e+06
324	1.408739e+06	-8.706487e+05	8.706487e+05	2.087107e+05	7.508398e+05	-3.882121e+05	2.679581e+06
325	1.459735e+06	-9.985177e+05	9.985177e+05	2.395322e+05	8.529336e+05	-4.606150e+05	2.743405e+06
326	1.508007e+06	-1.129819e+06	1.129819e+06	2.711383e+05	9.555421e+05	-5.384362e+05	2.796399e+06
327	1.553494e+06	-1.263726e+06	1.263726e+06	3.033030e+05	1.057788e+06	-6.213657e+05	2.838375e+06
328	1.596142e+06	-1.399405e+06	1.399405e+06	3.357969e+05	1.158812e+06	-7.090344e+05	2.869205e+06
329	1.635903e+06	-1.536020e+06	1.536020e+06	3.683894e+05	1.257783e+06	-8.010176e+05	2.888833e+06
330	1.672737e+06	-1.672737e+06	1.672737e+06	4.008504e+05	1.353900e+06	-8.968392e+05	2.897265e+06
331	1.706608e+06	-1.808729e+06	1.808729e+06	4.329523e+05	1.446404e+06	-9.959753e+05	2.894571e+06
332	1.737488e+06	-1.943182e+06	1.943182e+06	4.644714e+05	1.534578e+06	-1.097859e+06	2.880886e+06
333	1.765357e+06	-2.075301e+06	2.075301e+06	4.951901e+05	1.617754e+06	-1.201887e+06	2.856407e+06
334	1.790198e+06	-2.204312e+06	2.204312e+06	5.248986e+05	1.695321e+06	-1.307423e+06	2.821390e+06
335	1.812003e+06	-2.329466e+06	2.329466e+06	5.533959e+05	1.766725e+06	-1.413806e+06	2.776149e+06
336	1.830770e+06	-2.450048e+06	2.450048e+06	5.804921e+05	1.831474e+06	-1.520351e+06	2.721054e+06
337	1.846503e+06	-2.565377e+06	2.565377e+06	6.060093e+05	1.889140e+06	-1.626364e+06	2.656526e+06
338	1.859211e+06	-2.674810e+06	2.674810e+06	6.297827e+05	1.939364e+06	-1.731141e+06	2.583034e+06
339	1.868913e+06	-2.777746e+06	2.777746e+06	6.516624e+05	1.981853e+06	-1.833976e+06	2.501093e+06
340	1.875629e+06	-2.873631e+06	2.873631e+06	6.715139e+05	2.016385e+06	-1.934170e+06	2.411262e+06
341	1.879389e+06	-2.961957e+06	2.961957e+06	6.892188e+05	2.042810e+06	-2.031033e+06	2.314135e+06
342	1.880226e+06	-3.042269e+06	3.042269e+06	7.046761e+05	2.061043e+06	-2.123896e+06	2.210338e+06
343	1.878180e+06	-3.114163e+06	3.114163e+06	7.178022e+05	2.071072e+06	-2.212111e+06	2.100530e+06
344	1.873296e+06	-3.177290e+06	3.177290e+06	7.285315e+05	2.072950e+06	-2.295059e+06	1.985391e+06
345	1.865625e+06	-3.231357e+06	3.231357e+06	7.368167e+05	2.066795e+06	-2.372157e+06	1.865625e+06
346	1.855222e+06	-3.276127e+06	3.276127e+06	7.426287e+05	2.052789e+06	-2.442861e+06	1.741948e+06
347	1.842146e+06	-3.311420e+06	3.311420e+06	7.459565e+05	2.031170e+06	-2.506671e+06	1.615088e+06
348	1.826464e+06	-3.337116e+06	3.337116e+06	7.468073e+05	2.002236e+06	-2.563138e+06	1.485780e+06
349	1.808245e+06	-3.353150e+06	3.353150e+06	7.452055e+05	1.966333e+06	-2.611862e+06	1.354761e+06
350	1.787561e+06	-3.359515e+06	3.359515e+06	7.411928e+05	1.923855e+06	-2.652500e+06	1.222764e+06
351	1.764490e+06	-3.356260e+06	3.356260e+06	7.348270e+05	1.875240e+06	-2.684769e+06	1.090515e+06
352	1.739114e+06	-3.343488e+06	3.343488e+06	7.261816e+05	1.820962e+06	-2.708444e+06	9.587298e+05
353	1.711518e+06	-3.321356e+06	3.321356e+06	7.153446e+05	1.761528e+06	-2.723364e+06	8.281072e+05
354	1.681788e+06	-3.290073e+06	3.290073e+06	7.024178e+05	1.697471e+06	-2.729429e+06	6.993267e+05
355	1.650016e+06	-3.249896e+06	3.249896e+06	6.875152e+05	1.629345e+06	-2.726607e+06	5.730444e+05
356	1.616294e+06	-3.201130e+06	3.201130e+06	6.707624e+05	1.557721e+06	-2.714925e+06	4.498894e+05
357	1.580720e+06	-3.144122e+06	3.144122e+06	6.522946e+05	1.483179e+06	-2.694475e+06	3.304605e+05
358	1.543391e+06	-3.079262e+06	3.079262e+06	6.322561e+05	1.406305e+06	-2.665411e+06	2.153230e+05
359	1.504406e+06	-3.006978e+06	3.006978e+06	6.107983e+05	1.327683e+06	-2.627946e+06	1.050060e+05

deg	B (m/s)	F_lat signed (N)	|F_lat| (N)	F_lat x	F_lat y	F_lat z	F_parallel (N)
360	1.463866e+06	-2.927732e+06	2.927732e+06	5.880786e+05	1.247892e+06	-2.582353e+06	1.351950e-09

Appendix D. Source and variable ledger

Symbol	Meaning	Units/type
G_phase	fixed anisotropic traversal-cost matrix	dimensionless 3x3 SPD
c_hat	Earth-CMB traversal direction and outer axis	unit vector
e0	bar reference at phase zero	unit vector
A0	inner-axis reference at phase zero	unit vector
V_E	Earth velocity relative to K-Field/CMB frame	m/s
phi	outer phase omega_B t	radian
u	positive endpoint radial direction	unit vector
u_phi	du/dphi	dimensionless
B	u_phi^T G_phase V_E	m/s
C	u_phi^T G_phase u_phi	dimensionless
m	mass of each endpoint	kg
R	endpoint radius	m
omega_B	outer angular speed	rad/s
omega_A	inner angular speed=2omega_B	rad/s
kappa_F	constitutive traversal-force gain	native normalization
F_lateral	force perpendicular to c_hat	N per kappa_F
J	cycle lateral impulse	N s per kappa_F

Source ledger

#	Source	Use
1	K-Field_dual_axis_kinetic_mass_analysis_1784741897(1).json	authoritative apparatus, G_phase, Cell3 geometry, endpoint force law, and one-degree source data
2	K-Field_dual_axis_kinetic_mass_CMB_traversal_net_force_vector_1784751310.json	axial/lateral decomposition and cycle-direction reference
3	K-Field_dual_axis_CMB_traversal_correct_summary_1784754513.json	independent continuous extrema and corrected numerical values
4	K-Field_dual_axis_kinetic_mass_CMB_traversal_lateral_force_comptation_1784803951.json	complete machine-readable formula kernel, structural tables, phase ledger, and validation results

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